Incomplete Markets, Crop Portfolios, and Agricultural Productivity

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Using the Uganda National Panel Survey, I document that farmers face a trade-off between crop yields and risk, that most grow both high- and low-yield crops, and that poorer farmers concentrate on the low-yield, low-risk ones. I then develop an incomplete-markets model with a crop-portfolio choice, in which households trade off income returns against consumption smoothing. Wealth-poor households optimally under-invest in intermediates and shift toward safer crops. Crop portfolios, intermediate input use, and the wealth distribution are jointly determined in equilibrium. The model also reproduces the low pass-through of both income inequality and income volatility into consumption. Calibrated to rural Uganda, it implies that completing markets raises agricultural productivity by 12.8%, with 15.8% of the gain coming from eliminating the misallocation of inputs across farmers and crops. Consumption rises by 17.6% and welfare by 17.4%, with income inequality rising slightly and consumption inequality unchanged.

Keywords: Agriculture, Risk, Incomplete markets, Crop choice.

JEL Classification: O11, 013, E21.

Households in low-income countries live with high levels of risk while having limited access to formal financial markets or public safety nets. Many of these households rely heavily on agriculture, a sector characterized by both high income risk and particularly low productivity (Caselli, 2005; Restuccia, Yang, and Zhu, 2008; Gollin, Lagakos, and Waugh, 2014). In this article, I study how risk distorts agricultural decisions among heterogeneous farmers to then quantify the aggregate impact of completing financial markets in agriculture in Uganda.

Complete markets tests in developing countries typically show a high degree of consump-

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tion insurance (Townsend, 1994).[1] Yet informal insurance mechanisms are often imperfect and costly. When farmers in developing countries are offered insurance products, they tend to take riskier and more productive decisions, such as increasing fertilizer usage (Karlan, Osei, et al., 2014) or shifting to more productive crops (Mobarak and Rosenzweig, 2013; Cai, 2016). The absence of perfect insurance can have large aggregate effects, though knowledge on this dimension is still limited. Donovan (2021) shows how uninsurable risk reduces the intermediate share and in equilibrium creates a large agricultural productivity gap.

I use data from the Uganda National Panel Survey (UNPS), which provides detailed agricultural information at the household-plot level. I measure yields as the value of production—including non-sold output valued at local median consumption prices—per land unit ($/acre). Exploring the relationship between crop yields, risk, and crop choice, I find three empirical results. First, there is a strong positive correlation between crop yields and risk: crops such as plantain bananas, rice, and groundnuts offer high but volatile yields, while beans, cassava, and sorghum offer lower and less volatile yields. Second, the share of output from high-yield, high-risk crops rises along the wealth distribution: the poorest farmers obtain most of their production from low-yield crops (mainly beans, maize, and cassava), the richest mainly from high-yield crops (mainly plantain bananas). Third, most farmers cultivate high- and low-yield crops simultaneously. These patterns are robust across alternative risk measures, crop classifications, and data sources.

Motivated by these results, I develop an incomplete-markets model with a crop-portfolio choice. The economy is agrarian, populated by household farms that differ ex ante only in a permanent component of agricultural productivity. Households are infinitely lived with standard CRRA preferences, and income is endogenous to farming decisions: they choose intermediate inputs for both the high-return, high-risk technologies (high-yield crops) and the low-return, low-risk ones (low-yield crops), and also receive non-agricultural earnings from an exogenous stochastic process.[2] Financial markets are incomplete, in the tradition of Bewley (1986); Imrohoroğlu (1989); Huggett (1993); Aiyagari (1994); to capture the hardship of saving in developing countries, and following Donovan (2021) and Lagakos, Mobarak, and Waugh (2023), saving occurs through a risk-free asset that depreciates, subject to a no-borrowing constraint.[3] Because most agricultural production in Uganda is at subsistence level and rural labor markets are thin, I model the economy as a small open economy with fixed input prices and

[1] Townsend (1994) finds that household consumption responds little to idiosyncratic income shocks in rural India. Similarly, Paxson (1993) finds limited evidence that consumption tracks income seasonality in Thailand. More recent studies with results in the same direction include Chiappori et al. (2014), De Magalhães and Santaeulàlia-Llopis (2018), among others.

[2] Adamopoulos and Restuccia (2020) also model crop selection as a choice between two technologies, where the key trade-off is higher productivity versus lower fixed costs: a "cash-crop" technology and a "food-crop" technology.

[3] Although risk-sharing is a critical source of insurance in developing countries (Udry, 1994; Ligon, 1998; Fafchamps and Lund, 2003), for tractability insurance in the model is limited to self-insurance. Quantitatively, the model still captures the aggregate degree of insurance in the data—measured by the transmission of income to consumption volatility.

work in partial equilibrium.[4]

The model features two interlinked risk-management mechanisms: precautionary saving and precautionary income choices. When choosing intermediate inputs, households trade off higher agricultural returns against less consumption smoothing, and they adjust along two margins. The first is the total amount of intermediate use: holding the portfolio fixed, households raise returns by using more intermediates, and if only this margin operates there is a monotonic relationship between intermediate use, returns, and risk. The second margin is the portfolio choice across technologies: households can raise returns by shifting from low- to high-return technologies without increasing total intermediate use. Formally, this trade-off appears in the Euler conditions for intermediates, where a risk-induced wedge varies by technology. Portfolio choice thus adds an allocation problem within the household farm, so that the distribution of risk across technologies can generate misallocation. This microfounds the aggregate risk-induced wedge that an intermediate-use problem without a crop portfolio would otherwise leave unexplained.

Using the UNPS data, I calibrate the model to rural Uganda. Part of the ISA-LSMS project, the survey measures agriculture, consumption, income, and wealth within the same households, letting me reconstruct the entire budget constraint from a single dataset. Some parameters are set externally—the discount factor and risk aversion from the literature, and the crop-risk and non-agricultural income processes estimated directly from the data—while the remaining ones, the two technology factors, the dispersion of permanent productivity, and the depreciating return on saving, are calibrated within the model by a near-triangular method of moments, with each pinned mainly by one moment of the UNPS.

Beyond the moments it targets, the calibrated model reproduces the slope of the second empirical fact documented above: the rising share of high-yield, high-risk crops along the wealth distribution. It also delivers predictions for the joint distribution of consumption, income, and wealth. Because these are endogenous in an incomplete-markets economy, this class of models provide a theory of their joint distribution—used mostly to account for inequality in the United States and other high-income economies (Castañeda, Díaz-Giménez, and Ríos-Rull, 2003; Krueger and Perri, 2006). For Sub-Saharan Africa, De Magalhães and Santaeulàlia-Llopis (2018) document two features of the region: a low transmission of income inequality to wealth inequality (low accumulation in the life cycle), and a low transmission of income inequality to consumption inequality (high consumption insurance). Yet, to my knowledge, no structural account of these distributions exists for the region. This paper provides one. Although the calibration targets no dispersion moment, the model reproduces the second fact, high-insurance, in both dimensions in which it appears in the data: a cross-sectional one—the low pass-through of income inequality to consumption inequality—and a dynamic one—the low pass-through of idiosyncratic income volatility to consumption volatility. It thus captures quantitatively the high consumption insurance observed in rural developing economies and matches the

[4]This abstracts from the labor-reallocation channel Donovan (2021) finds important in India—which would raise the gains—and from local relative-price adjustment—which could lower them.

broader consumption–income distribution of rural Uganda with standard preferences and no consumption target, though it under-predicts the concentration of liquid wealth.This suggests that the incomplete-markets toolkit—the workhorse for inequality and welfare analysis in rich economies—is a useful and underused framework in developing economies.

Next, I develop a planner's problem that replicates the competitive-equilibrium allocations of intermediates and consumption under complete markets in agriculture, where households maximize discounted expected profits through their crop-portfolio choices. Comparing the calibrated benchmark with this counterfactual delivers the main results.

Completing financial markets raises agricultural productivity by 12.8%; because labor and land are fixed, output and productivity—per worker and per land unit—rise together. The gain comes from two channels: a 29.0% rise in average intermediate use, and a more efficient allocation of intermediates across the crop portfolios of heterogeneous farmers. Holding aggregate intermediate use at its benchmark level and allocating it efficiently accounts for 15.8% of the total increase in production, 12.2% across farmers and 3.2% across crops. The crop portfolio itself accounts for 16.0% of the gain: an economy with a single technology, recalibrated to the same average agricultural output and the same volatility of total agricultural output, delivers a productivity gain of 10.7% against 12.8% here. Completing markets also lowers the demand for costly precautionary saving which, together with the productivity gain, raises aggregate consumption by 17.6%.

In this model, the effect on consumption inequality is theoretically ambiguous: completing markets removes the dependence of consumption on agricultural shocks (an egalitarian force), but the largest output gains accrue to high-productivity households concentrated in the middle and upper distribution, and the reduction in costly saving benefits the top most. After completing markets, income inequality rises while consumption inequality remains almost unchanged, as the egalitarian and inequality-enhancing forces roughly offset.[5] In consumption-equivalent terms, households would need a 17.4% increase in lifetime consumption to be indifferent between being born into the incomplete-markets economy and the economy with complete agricultural markets.

Related Literature

The closest paper is Donovan (2021), who shows that, in equilibrium, uninsurable risk lowers the intermediate share, so that completing financial markets raises agricultural productivity and narrows the cross-country productivity gap. I contribute in three ways. First, risk distorts intermediate use along two margins: its level, as in his model, and its allocation across crops, which in turn feeds back on the level. Second, I quantify the welfare gains from completing agricultural insurance: the gains in aggregate consumption are larger than in output, 17.6% against 12.8%, and the distributional effects are non-trivial. Third, in a poorer,

[5]Households with low permanent components—limited agricultural skills, small landholdings, or remote, unproductive locations—may gain little from insurance.

subsistence-based economy and under different assumptions—a permanent productivity component, no subsistence requirement, and no cross-sector labor reallocation—the core insight survives: uninsurable risk lowers agricultural productivity.[6]

More broadly, the paper contributes to two strands of work on agricultural productivity in poor countries: a macro-development literature on misallocation, and a micro literature on risk in farm decisions. The first documents the large productivity gap between rich and poor countries and the barriers behind it: high transport costs and subsistence (Gollin and Rogerson, 2014), land-market frictions under communal tenure (Gottlieb and Grobovšek, 2019; Adamopoulos and Restuccia, 2020; Chen, Restuccia, and Santaeulàlia-Llopis, 2023), land and financial frictions (Manysheva, 2022), land-use restrictions that distort crop choice (Le, 2020), among others. In that literature, misallocation operates largely across farmers, in the factors they use (Adamopoulos and Restuccia, 2014). I add the margin misallocation across crops, within the farm. It connects to Adamopoulos and Restuccia (2022), who show that the productivity gap reflects economic choices—such as which crops are grown, and how—rather than unfavorable geography; I show that part of that inefficient crop allocation stems from optimal choices of farmers facing uninsurable risk. The second examines the role of risk in agricultural decisions: Dercon (1996) shows that risk distorts crop choices, Dercon and Christiaensen (2011) that it distorts fertilizer use, and Rosenzweig and Binswanger (1993) that its effect on investment depends on wealth, while Kurosaki and Fafchamps (2002) find that crop choices respond to risk even when accounting for village-level insurance. The contribution of this paper is twofold: it develops a framework that incorporates these micro-level mechanisms into an equilibrium setting with an endogenous wealth distribution, and it quantifies the aggregate consequences of them for agricultural productivity and welfare. Finally, by quantifying the productivity and welfare cost of the distortions households accept to smooth consumption, the paper speaks to work documenting a trade-off between consumption smoothing and growth, at the micro (Rosenzweig and Wolpin, 1993; Morduch, 1995) and macro (Munshi and Rosenzweig, 2016; Santaeulalia-Llopis and Zheng, 2018) levels.

1 EMPIRICAL FINDINGS

This section explores the relationship between agricultural yields, risk, and farmers' crop choices across the main crops grown in Uganda. I document three facts: (1) there is a positive yield–risk tradeoff across crops, robust across alternative risk measures and data sources; (2) low-yield, low-risk crops are the main share of output for poorer farmers; and (3) most farmers grow both low- and high-yield crops at the same time.

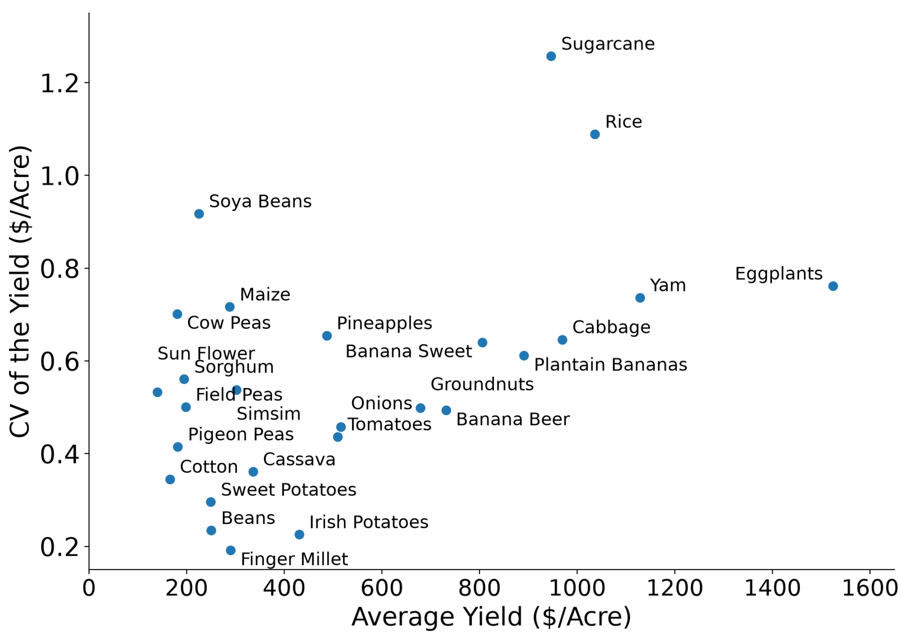
To document these facts, I mainly use the first five waves of the Uganda National Panel

[6] Donovan (2021) studies India with a two-sector model (agriculture and a riskless industry); completing markets there raises productivity through a higher intermediate share, reduced misallocation across farmers, and labor reallocation to industry. I omit the third channel, which requires an industrial sector to absorb post-shock farm labor—hard to justify in rural Uganda.

Survey (UNPS) and compute crop yields, defined as the value of production per land unit ($/acre). The value of production is the household-reported value of crop sales plus non-sold production, valued at district-level median consumption prices, and land is the reported acreage devoted to the crop.

Fact 1: a yield–risk tradeoff. Figure 1 plots average monetary yield against the coefficient of variation (CV) of yield across crops, with observations aggregated at the wave–season level so that dispersion reflects variation over time. Higher-yielding crops are markedly more volatile. Crops such as plantain bananas (matooke) reach average yields of about $891 per acre with a CV of 0.61, whereas crops such as beans yield around $250 per acre with a CV of 0.24—a threefold difference in yields and a two-and-a-half-fold difference in risk. The positive relationship is strong with a correlation of 0.68 and holds across alternative risk measures and data sources—which I discuss in section 1.1.

Figure 1: Crop Yield and Crop Risk in Uganda

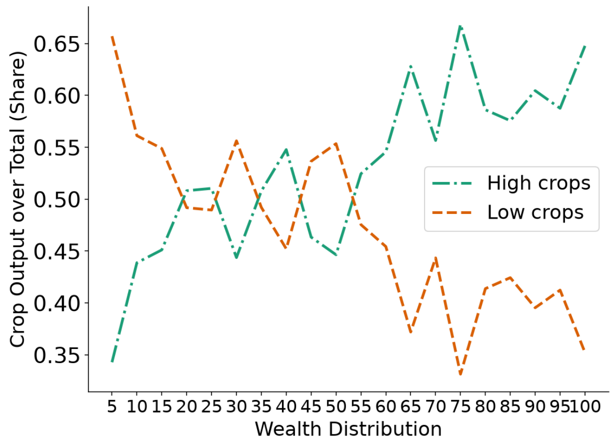


Notes: Yield observations ($/acre) at the wave level. Crops with a planting-to-harvest cycle under two years. *Source:* UNPS 2009/10–2015/16.

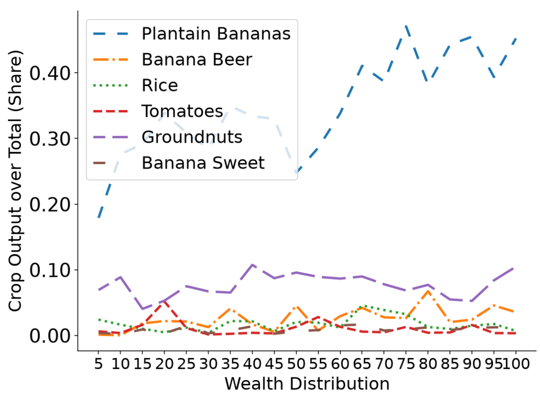
Fact 2: poorer farmers concentrate output in low-yield crops. I group the non-long-term crops into high- and low-yield categories by whether their average yield is above or below the median crop. High-yield crops include bananas (beer), plantain bananas, sweet bananas, cabbage, eggplants, groundnuts, Irish potatoes, onions, pineapples, pumpkins, rice, sugarcane, tomatoes, and yam; low-yield crops include beans, cassava, cotton, dodo, cowpeas, field peas, finger millet, maize, pigeon peas, simsim, sorghum, soya beans, sunflower, and sweet potatoes. Figure 2 shows the crop portfolio along the wealth distribution. Low-yield crops make up the majority of poorer farmers' output and high-yield crops the majority of richer farmers. Among the poorest farmers (0–10th percentiles) low-yield crops are around 60% of production; in the middle (15th–55th) output is split roughly evenly; and at the top (60th percentile and above) high-yield crops account for about 60%.

Figure 2: Crops Production along the Wealth Distribution

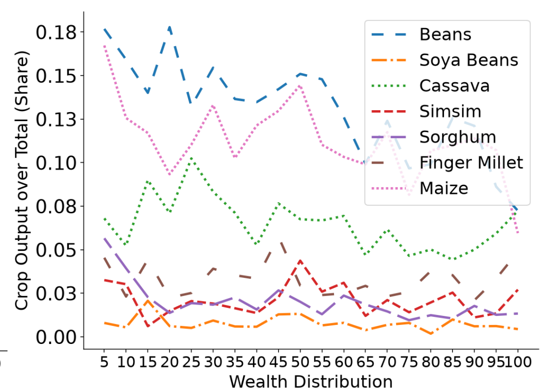
(a) High-Yield vs Low-Yield Crops



(b) High-Yield Crops



(c) Low-Yield Crops



Notes: Wealth includes estimated land value, household assets, livestock, and farming capital. Crop production and wealth ranking computed from the average across the five waves. *Source:* UNPS 2009/10–2015/16.

Sub-plots (2b) and (2c) show the six most cultivated high-yield and seven most cultivated low-yield crops. The rise in the high-yield share is driven mainly by plantain bananas, which go from 33% of total production for the bottom 50% to 40% for the top 35%; conversely, beans and maize together fall from about 30% of the bottom 50% poor's output to under 20% for the rich. The same gradient appears more strongly along the income distribution (Figure 6 in the Appendix).

Fact 3: farmers grow both types. Most farmers grow both low- and high-yield crops. Table 1 reports the share of households growing low-yield crops, high-yield crops, and both, across waves. On average 65.0% grew high-yield crops and 96.2% grew low-yield crops; over the whole panel, 98.6% grew low-yield crops, 76.1% grew high-yield crops, and 75.3% grew both.

Table 1: Share of Farmers Cultivating Crops

	High Yield (%)	Low Yield (%)	Both (%)
2009–10	61.99	93.29	60.75
2010–11	68.34	97.25	66.87
2011–12	68.68	97.39	66.92
2013–14	61.19	96.26	59.28
2015–16	64.83	97.01	62.08
Mean	65.01	96.24	63.18
Any wave	76.14	98.64	75.27

Notes: Mean denotes the average of waves' averages. Any wave is the share of farmers growing the crops in at least one of the five waves. *Source:* UNPS 2009/10–2015/16.

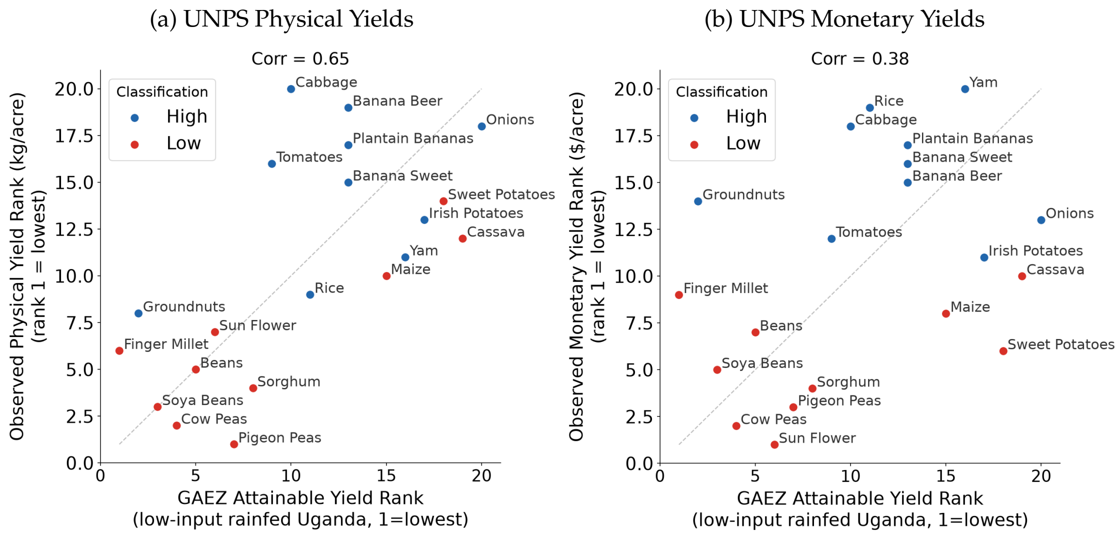
1.1 Robustness

I now show that the crop ranking and the yield–risk tradeoff are robust: that the ranking reflects underlying yields rather than input intensity, that the tradeoff holds across risk measures, that risk is driven mainly by physical production risk, that it does not rest on a single crop, and that it survives alternative classifications.

Crop yield ranking and GAEZ data. A first concern is whether the monetary yield ranking reflects how intensively crops are cultivated rather than their underlying yields.[7] I address it in two ways. First, I compare the UNPS ranking with the FAO Global Agro Ecological Zones (GAEZ) attainable yields for Uganda (1981–2010) under low-input rainfed conditions. Because these are based on soil and climate alone, they are independent of farmers' choices. Figure 3 compares the GAEZ ranking with the UNPS physical (3a) and monetary (3b) rankings. GAEZ yields correlate strongly with UNPS physical yields (0.65): both place bananas, onions, yam, Irish potatoes, sweet potatoes, and rice as high-yield crops, and finger millet, sunflower, sorghum, beans, and peas as low. The correlation with the monetary ranking is weaker (0.38), and the gap is prices: cassava and sweet potatoes rank low in value but high in physical yield because their prices are relatively low, while groundnuts and, to some extent, rice are the reverse, ranking high in value but low in physical. Second, I re-rank crops by net yield, subtracting the value of intermediate inputs currently used, so that the ordering reflects yields net of input intensity. The classification is almost unchanged: every crop keeps its category except three minor high-yield crops (pumpkins, eggplants, and pineapples), which move down, and cassava, which moves up to high yield. The ranking is therefore largely agronomic, with the value differences for maize, cassava, sweet potatoes, rice, and groundnuts driven by prices; only cassava and maize are crops whose classification is less clear.

[7] This matters because the high/low classification used in the model and counterfactuals would be endogenous to input use if the ranking reflected how intensively crops are cultivated, in which case the quantitative experiment could reshuffle the crop categories.

Figure 3: Crop Ranking: GAEZ Attainable Yields vs. UNPS Observed Yields



Notes: X-axis ranks crops from GAEZ attainable yields for Uganda; Y-axes rank UNPS observed physical (kg/acre, panel a) and monetary ($/acre, panel b) yields. Corr is the Spearman correlation. *Source:* FAO-GAEZ (v4); UNPS 2009/10–2015/16.

Measuring crop risk. Appendix Table 8 reports, for each crop, the average yield and five measures of dispersion, from wave-level CVs to within-household fixed-effects-residual measures, and at the district level. The correlation between yield and risk is positive under every measure, from 0.37 to 0.68 across the value-based measures. To address classical recall measurement error and spatially correlated weather shocks, I recompute the relationship at the district level, where averaging households within each district–wave nets out classical measurement error; the trade-off is not weakened but is stronger (0.68), so it is not an artifact of classical measurement error. A second concern is that, because yield is value per acre, the pattern could reflect price rather than physical variation. It does not: the trade-off survives in purely physical terms (0.22)—attenuated relative to the value measure (0.44), since physical yields discard the price dimension, but with the sign and ordering preserved. A variance decomposition confirms this: using the exact identity $\log(\text{value}/A) = \log(\text{kg}/A) + \log p$ and valuing all output at the median district consumption price, physical quantity accounts for about three-quarters of the household idiosyncratic yield variance (0.78) and the price of valuation for under a third (0.28), with a small negative covariance; the pattern holds crop by crop. The trade-off appears to be primarily driven by physical production risk.[8]

[8] The decomposition is object-dependent. For the variance of crop-wave means—the object behind the scatter in Figure 1—price appears to dominate (share 0.89), but this is an aggregation artifact: averaging over households in a crop-wave removes idiosyncratic physical shocks, while the common district price is not averaged down. Applied to the risk households actually face—the within-crop variance of household yields net of wave and season effects, which is what the model and calibration use—quantity dominates (0.78; price 0.28; covariance -0.06), and adding household fixed effects leaves this essentially unchanged (0.77; 0.31). The result is robust to how output is valued: the identity requires valuing all output at the district price, whereas the baseline blends realized sale value for sold output with the district price for the rest; re-running on the blended measure moves the quantity share by less

Perennial crops and banana risk. The analysis retains crops with a planting-to-harvest cycle under two years and excludes long-gestation perennials (coffee, tea, tobacco, and others), whose yields and volatility are hard to measure and whose fixed costs are a margin this paper does not study. Among the retained perennial crops, only bananas are large enough to shape the aggregate patterns: they are the high-yield crop that accounts for most of the wealth gradient in Fact 2. Because bananas are perennial, one might worry that their measured volatility reflects the maturation of young plants rather than genuine production risk. I check this by comparing growers first observed cultivating bananas in the early waves ("established") with later adopters.[9] If maturation were inflating banana risk, established growers would show lower volatility. They do not: over the common window their within-household CV is 0.79, slightly above later adopters (0.71) and the cross-crop median (0.72) while over the full panel the gap is wider, 0.88 against 0.71 (Appendix Table 11). Bananas are also a staple grown by many Ugandan farmers rather than a specialized crop and recalibrating the model without bananas delivers gains close to those of the baseline classification.

Alternative crop classifications. The trade-off between high- and low-return crops is preserved across alternative classifications. Appendix Table 12 reruns it five ways: keeping only crops with more than 150 growers, cutting at terciles rather than the median, excluding bananas, using the GAEZ physical ranking, and grouping crops and measuring their risk at the district level. In every case the group of high yield crops exhibits higher risk than the low yield one. Across the value-based splits the variance ratio stays around its baseline value of 1.72 (from 1.69 to 1.79), rises to 2.10 when crops are grouped at the district level, while it decreases to 1.08 in the GAEZ split. This is expected since GAEZ ranks crops by agronomic physical potential (attainable yield), not by return; and because it contains no time variation, the risk it is paired with comes from the UNPS panel. Finally, The two-group structure is not discarding much of the crop-level variation in risk. Partitioning the same crops into three or five groups raises the average within-group output risk by only $3-8\%$ and leaves the spread of risk across groups essentially unchanged.

2 Model

2.1 Primitives of the Model

The model depicts an agrarian economy populated by a continuum of household farms—or simply households—of measure one facing idiosyncratic risk with no aggregate risk. They are ex-ante identical except for a permanent component in agricultural production.

than three percentage points. The small negative covariance is the natural-insurance association of Allen and Atkin (2022), concentrated at the market rather than the idiosyncratic level (-0.65 across waves versus -0.06 to -0.08 idiosyncratically). Appendix Tables 9 and 10 report the full results.

[9] Plant age is not observed, so I use a proxy: farmers first observed cultivating bananas in 2009/10 or 2010/11 ("established") plausibly have fewer young plants than farmers first observed growing cultivating from 2011/12 onward. I compute the risk measures for both groups over the common window 2011/12–2015/16, in which both are observed, so the comparison spans the same years.

Preferences. Households live infinitely and maximize expected utility

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t u(c_t)\right] \qquad\qquad u(c) = \frac{c^{1-\rho}}{1-\rho}$$

where $\beta \in (0,1)$ is the discount factor, $\mathbb{E}_0$ denotes the expectation at time 0, and ρ is the coefficient of relative risk aversion.

Agricultural production. Households generate farming income by choosing intermediate inputs for two technologies—high-return, high-risk and low-return, low-risk—that produce the same single consumption good. Land and labor are fixed, so intermediate inputs are the only production choice, and they are committed before shocks are realized. Both technologies exhibit decreasing returns to scale ($\gamma, \alpha \in (0,1)$). The production functions take the following form

$$y^h_{it} = \theta_{it} z_i A (m^h_{it-1})^{\alpha} \tag{1}$$
$$y^l_{it} = \varepsilon_{it} z_i B (m^l_{it-1})^{\gamma} \tag{2}$$

in which m^h_{it-1} and m^l_{it-1} are the intermediate input choices in high-return and low-return technologies, respectively, for household i at period $t-1$. γ and α are the elasticities of intermediates on output, A, B are technology-neutral productivity factors, while y^h_{it} represents the output of the high-return technology and y^l_{it} from the low-return technology. z_i is the household-specific component which captures permanent or quasi-permanent characteristics in agricultural production including differences in productivity as well as location and land characteristics.[10] Shocks θ_{it} and ε_{it} are correlated across technologies ($\sigma_{\theta\varepsilon} > 0$), independent across households i and time t, and follow a bivariate log-normal distribution

$$(\ln\theta_{it},\ \ln\varepsilon_{it}) \sim \mathcal{N}(\bar{\mu}, \Sigma), \qquad \bar{\mu} = \begin{bmatrix} -\sigma^2_{\theta}/2 \\ -\sigma^2_{\varepsilon}/2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma^2_{\theta} & \sigma_{\theta\varepsilon} \\ \sigma_{\theta\varepsilon} & \sigma^2_{\varepsilon} \end{bmatrix},$$

normalized so that θ_{it} and ε_{it} have mean one.

Non-agricultural earnings. Households also receive non-agricultural earnings from an AR(1) process $y_{na,it}$,

$$\ln y_{na,it+1} = b + \rho_{na} \ln y_{na,it} + u_{it}, \qquad u_{it} \sim \mathcal{N}(0, \sigma^2_{na}),$$

where ρ_{na} is the autocorrelation coefficient and u_{it} the white-noise innovation. Non-agricultural earnings are about a third of household income in rural Uganda and are received by 72% of households—a heterogeneous mix of self-employment, casual wage work, and livestock. I model these earnings as an exogenous process, independent of agricultural shocks.[11]

[10] The permanent component is a central assumption in this work. Farmers in less productive regions (e.g., semi-arid northern Uganda) face different crop choices and lower average productivity, and less-skilled farmers produce less. Modeling these fixed or quasi-fixed characteristics parsimoniously lets the model (i) separate risk-induced from structural dispersion in output, (ii) generate a non-degenerate consumption distribution even under complete markets, and (iii) study heterogeneous effects of completing markets. As land is not modeled explicitly, z_i also absorbs land heterogeneity, natural where land markets are rigid or absent.

[11] The correlation of agricultural and non-agricultural income is 0.083 (0.016 once household and time effects are

Financial market. Despite limited access to formal financial markets, households smooth consumption through informal risk-coping mechanisms—risk-sharing networks (e.g., (Udry, 1994; Ligon, Thomas, and Worrall, 2002)) and precautionary savings. In poor countries they accumulate assets such as crop storage (Fafchamps, Udry, and Czukas, 1998; Kazianga and Udry, 2006), livestock (Rosenzweig and Wolpin, 1993), and other holdings (Collins et al., 2009) to cope with adversity. I capture this insurance tractably through an exogenously incomplete financial market: households save in a single risk-free asset a that depreciates at rate δ, and cannot borrow. The depreciation rate captures the finding that saving in poor countries is costly. This structure follows Donovan (2021) and Lagakos, Mobarak, and Waugh (2023), and is consistent with evidence that exogenously incomplete, saving-only markets fit developing rural data (in Thailand) better than risk-sharing alternatives (Karaivanov and Townsend, 2014).

2.2 The Household's Problem

Consider the household's problem under stationarity. Households maximize lifetime utility by choosing consumption, savings, and intermediate investment in the two technologies each period. The Bellman equation defines the problem in recursive form. I define the state space by three state variables: x, z, and y_{na}. Variable x represents cash-on-hand, the sum of agricultural income and the depreciated risk-free asset holdings. z is the household-specific permanent component in agriculture, and y_{na} is the current realization of the non-agricultural earnings process. Let $V(x, z, y_{na})$ be the value function at state (x, z, y_{na}); the Bellman equation is

$$V(x, z, y_{na}) = \underset{a', c, m'_h, m'_l}{\text{Max}} u(c) + \beta \mathbb{E}\left[V(x', z, y'_{na})\right] \tag{3}$$

subject to

$$a' \geq 0 \tag{4}$$

$$c(x, z, y_{na}) + a'(x, z, y_{na}) + pm'_h(x, z, y_{na}) + pm'_l(x, z, y_{na}) = x + y_{na} \tag{5}$$

$$x' = \theta' z A \left(m'_h(x, z, y_{na})\right)^{\alpha} + \varepsilon' z B \left(m'_l(x, z, y_{na})\right)^{\gamma} + (1 - \delta) a'(x, z, y_{na}) \tag{6}$$

$$(\ln \theta', \ln \varepsilon') \sim \mathcal{N}(\bar{\mu}, \Sigma) \tag{7}$$

$$\ln y'_{na} = b + \rho_{na} \ln y_{na} + u, \qquad u \sim \mathcal{N}(0, \sigma^2_{na}) \tag{8}$$

in which c, a', m'_h, and m'_l are the policy functions that solve the household-farm problem in state (x, z, y_{na}). To reduce notation, current-period variables (x) carry no subscript t, and next-period variables are denoted with a prime (x'). At the optimum, the first-order conditions set the derivative of the maximand $u(c) + \beta \mathbb{E}[V(x')]$ with respect to each choice (c, a', m'_h, m'_l) to

removed).

zero. Combining these with the envelope condition, implies the following first order conditions

$$u_c\left(c(x,z,y_{na})\right) \geq \beta(1-\delta)\mathbb{E}\left[u_c\left(c(x',z,y'_{na})\right)\right] \tag{9}$$

$$pu_c\left(c(x,z,y_{na})\right) = \beta\mathbb{E}\left[u_c\left(c(x',z,y'_{na})\right)\theta' zA\alpha\left(m'_h(x,z,y_{na})\right)^{\alpha-1}\right] \tag{10}$$

$$pu_c\left(c(x,z,y_{na})\right) = \beta\mathbb{E}\left[u_c\left(c(x',z,y'_{na})\right)\varepsilon' zB\gamma\left(m'_l(x,z,y_{na})\right)^{\gamma-1}\right] \tag{11}$$

and the budget constraint is

$$c(x,z,y_{na}) = x + y_{na} - a'(x,z,y_{na}) - pm'_h(x,z,y_{na}) - pm'_l(x,z,y_{na}). \tag{12}$$

The solution to the household problem is the set of policy functions that solve the system of equations (9), (10), (11), and (12).

To see the implications of risk for household choices, first note that equations (10) and (11) can be rewritten as

$$p = \frac{\mathbb{E}\left[u_c\left(c(x',z,y'_{na})\right)\theta'\right]}{u_c\left(c(x,z,y_{na})\right)}\beta zA\alpha\left(m'_h(x,z,y_{na})\right)^{\alpha-1} \tag{13}$$

$$p = \frac{\mathbb{E}\left[u_c\left(c(x',z,y'_{na})\right)\varepsilon'\right]}{u_c\left(c(x,z,y_{na})\right)}\beta zB\gamma\left(m'_l(x,z,y_{na})\right)^{\gamma-1} \tag{14}$$

in which p is the marginal cost and $zA\alpha(m'_h(x,z,y_{na}))^{\alpha-1}$ and $zB\gamma\left(m'_l(x,z,y_{na})\right)^{\gamma-1}$ are the marginal products of m_h and m_l, respectively. Suppose households can perfectly smooth consumption across states. Then the ratio of marginal utilities equals one, and optimal m_h and m_l are such that the discounted expected marginal returns equal the price. That is the expected-profit-maximization scenario.[12] Nevertheless, the lack of perfect consumption smoothing creates a risk-induced wedge between marginal cost and marginal products. Specifically, when the marginal utility of current consumption exceeds the expected marginal utility of next period, adjusted for shocks, the ratio of marginal utilities is less than one. Consequently, marginal products must surpass marginal cost. Given the production functions' decreasing returns to scale, a higher marginal product corresponds to lower investment in intermediates (m_h, m_l). This is the first margin of adjustment households use to mitigate risk exposure.

The second margin of adjustment is the technology portfolio. Risk not only distorts intermediate input levels but also their allocation across the two technologies. Dividing (13) by (14), the household's optimal intermediate input choices can be rewritten as

$$B\gamma\left(m'_l(x,z,y_{na})\right)^{\gamma-1} = \frac{\mathbb{E}\left[\theta' u_c\left(c(x',z,y'_{na})\right)\right]}{\mathbb{E}\left[\varepsilon' u_c\left(c(x',z,y'_{na})\right)\right]}A\alpha\left(m'_h(x,z,y_{na})\right)^{\alpha-1}. \tag{17}$$

[12] Expected discounted profit maximization implies that marginal cost equals expected discounted marginal return:

$$p = \beta\mathbb{E}_\theta\left[\theta' zA\alpha m_h^{\alpha-1}\right] = \beta zA\alpha m_h^{\alpha-1} \tag{15}$$

$$p = \beta\mathbb{E}_\varepsilon\left[\varepsilon' zB\gamma m_l^{\gamma-1}\right] = \beta zB\gamma m_l^{\gamma-1} \tag{16}$$

The absence of perfect consumption smoothing introduces a risk-induced wedge across the marginal products of the two technologies. From equation (17), note first that marginal utility decreases with consumption while consumption increases in the shock realizations. Since the shocks in the high technology have greater dispersion than in the low technology, $\sigma_\theta > \sigma_\varepsilon$, the low realizations of ε are larger than the low realizations of θ, and the high realizations of ε are smaller than the high realizations of θ. Because marginal utility is decreasing in the shock realization, the expectation in the denominator is larger than the expectation in the numerator. Hence marginal returns in the high technology exceed those in the low technology whenever consumption is not uniform across states. This is the second margin of adjustment, in which households optimally shift investment across the technology portfolio to mitigate risk exposure.

The absence of perfect consumption smoothing drives a risk wedge between the marginal products of the two technologies. In equation (17), next-period marginal utility $u_c(c')$ is high precisely when consumption—and hence the shock—is low. Because the high technology's shock is more dispersed ($\sigma_\theta > \sigma_\varepsilon$), θ' covaries more strongly and negatively with $u_c(c')$ than ε' does: bad draws in the high technology hit consumption harder. Hence $\mathbb{E}[\theta' u_c(c')] < \mathbb{E}[\varepsilon' u_c(c')]$, so the risk-adjusted marginal product required in the high technology exceeds that in the low technology whenever consumption is not smooth. This is the second margin of adjustment: households tilt intermediate investment away from the high-risk technology toward the low-risk one.

Figure 4 plots the value and policy functions from the household problem at the calibrated parameters. For clarity, the figure shows two representative levels of each state—a lower and a higher value of the permanent component z and of non-agricultural earnings y_{na}—out of the five used in the calibration (Section 4). The horizontal axis is cash-on-hand x; color and markers denote the permanent component (red, high z; blue, low z), while solid lines denote low y_{na} and dashed denote high y_{na}. For a given cash-on-hand level x, households with a high permanent component hold fewer risk-free assets (4b) and consume more (4c) than households with the lower component; likewise, households with higher non-agricultural earnings y_{na} save less and consume more.

Sub-plots (4d) and (4e) show the policy functions $m'_\ell(x, z, y_{na})$ and $m'_h(x, z, y_{na})$; the horizontal lines mark, for each permanent component z, the input level that maximizes discounted expected profits—the complete-markets allocation. Together they show how uninsurable risk shapes intermediate investment across heterogeneous farmers. First, poor households—those with low cash-on-hand x—under-invest in intermediates in both technologies (the first margin of adjustment). Second, as households become richer, intermediate investment rises toward the profit-maximizing level set by their permanent component. Third, because saving in the risk-free asset is costly in this economy, the richest households over-invest in intermediates, especially in the low-return technology.

To show how uninsurable risk distorts the technology portfolio (the second margin of adjustment), sub-plot (4f) plots the ratio of the expected marginal products of the two technologies across the state space. By equation (17), if households could perfectly smooth consumption across states, this ratio would equal one, and farm and household decisions would separate. Under incomplete markets, instead, the ratio exceeds one. The poorer the household and the higher its permanent component, the further the ratio rises above one—and the larger the under-investment in the high- relative to the low-return technology.

In line with the results of Rosenzweig and Binswanger (1993), in this economy the production choices of poor farmers are more distorted by the presence of risk. The model shows that even in the absence of fixed costs, the poor might not invest enough to move to more productive technologies, given their limited ability to cope with risk. Thus, poverty quasi-traps might exist even without fixed costs—a conclusion similar to Dercon and Christiaensen (2011).

2.3 Equilibria

Rural Uganda is an economy based on subsistence farming. Among all crops cultivated in Uganda, farmers devote only 26% of their production to the market—25% for high crops and 21% for low crops—and only 8% of farm labor is hired.[13] Most of what these households produce in agriculture and almost all of the labor they use never reach a market. Domestic market clearing is therefore not the right description of how quantities are determined here, and I model rural Uganda as a small open economy with fixed prices.

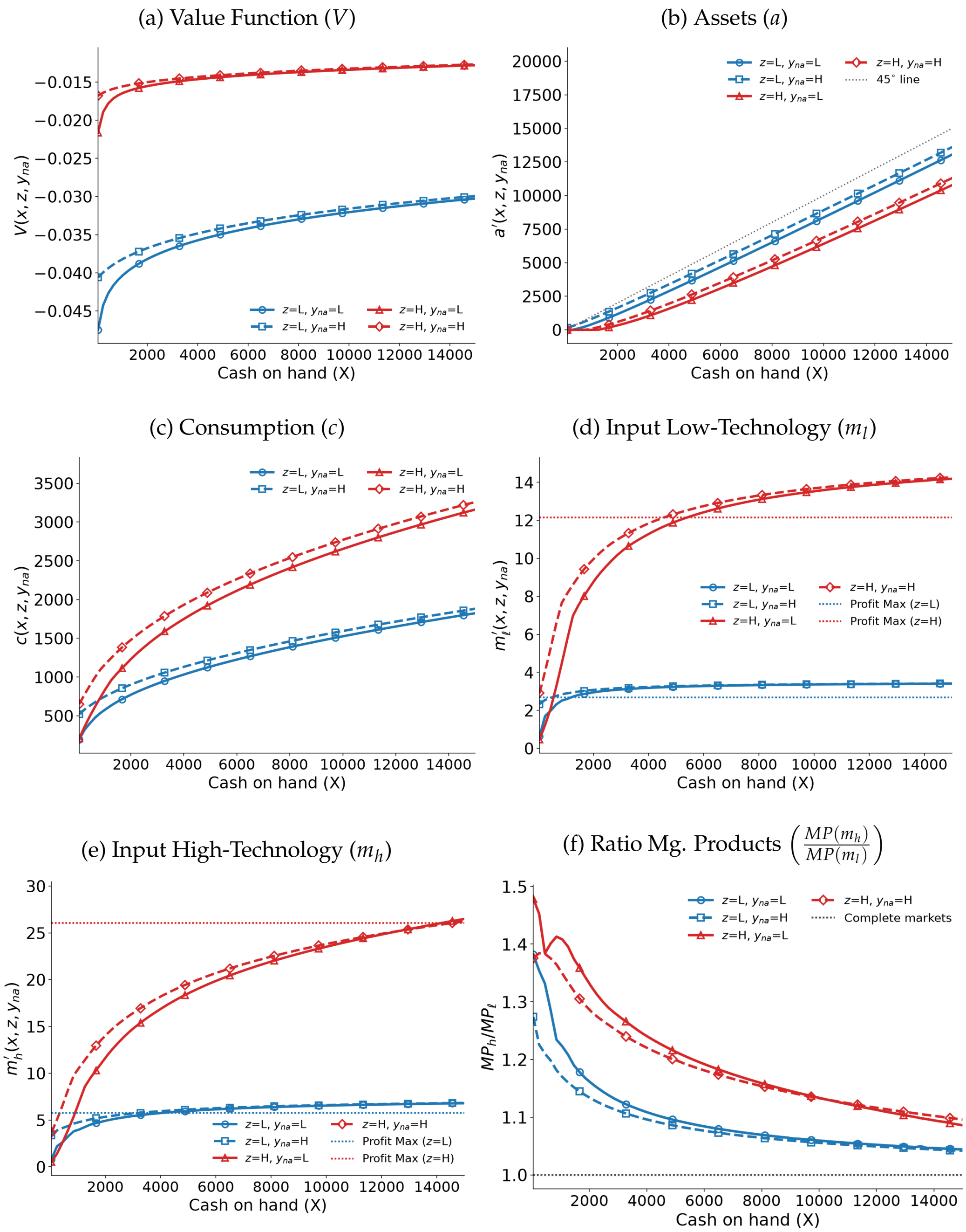
The two crop technologies and non-agricultural earnings are denominated in a single consumption good, so the economy has one resource constraint and no endogenous relative price across technologies. Labor is supplied within the household and fixed: there is no labor market and no occupational choice between agriculture and non-agriculture. Intermediate inputs are a household choice and their price is exogenous—the costly modern inputs (chemical fertilizer, pesticides) are imported, so farmers are price-takers, while the rest (saved seed, manure) are largely self-produced. The return on savings is likewise exogenous, reflecting the limited reach of formal finance.

Stationary Recursive Competitive Equilibria. The stationary RCE is defined by a stationary distribution $\lambda(x,z,y_{na}) = \lambda^*(x,z,y_{na})$, a transition function $Q\big((x',z,y'_{na}),(x,z,y_{na})\big)$, a value function $V(x,z,y_{na})$, and a set of policy functions $g(x,z,y_{na}) := \{c(x,z,y_{na}), a'(x,z,y_{na}), m'_h(x,z,y_{na}), m'_l(x,z,y_{na})\}$ such that:

1. Policy functions $g(x,z,y_{na})$ solve the household problem and $V(x,z,y_{na})$ is the associated value function.

[13] Table 13 in the Appendix reports the mean across households of the share of a household's production allocated to the market and to non-market uses (own consumption, storage, gifts), together with the share of farmers devoting a positive amount to each use. Table 15 reports farm labor use, separating household from hired supply.

Figure 4: Household's Optimality: Policy Functions



Notes: Value and policy functions depend on the state cash-on-hand x (x-axis), the permanent component z (red and blue), and the realization of non-agricultural earnings y_{na} (solid and dashed lines). Dotted lines in (d), (e), and (f) represent the expected-profit-maximization (complete-markets) allocations.

2. Given policy functions $g(x,z,y_{na})$, the aggregate resource constraint holds:

$$\begin{aligned}\int_{\mathcal{S}} c(x,z,y_{na})d\lambda^*(x,z,y_{na}) = &\int_{\mathcal{S}} \left[\theta z A(m_h'(x,z,y_{na}))^{\alpha} + \varepsilon z B\left(m_l'(x,z,y_{na})\right)^{\gamma}\right] d\lambda^*(x,z,y_{na}) - \\ &- \int_{\mathcal{S}} \left[p m_h'(x,z,y_{na}) + p m_l'(x,z,y_{na})\right] d\lambda^*(x,z,y_{na}) + \\ &+ \int_{\mathcal{S}} y_{na} d\lambda^*(x,z,y_{na}) - \delta \int_{\mathcal{S}} a'(x,z,y_{na}) d\lambda^*(x,z,y_{na})\end{aligned}$$

3. The stationary distribution $\lambda^*(x,z,y_{na})$ associated with $g(x,z,y_{na})$ satisfies the law of motion

$$\lambda^*(x',z,y_{na}') = \int_{\mathcal{S}} Q\left((x',z,y_{na}'),(x,z,y_{na})\right) d\lambda^*(x,z,y_{na})$$

.[14]

3 Data: Uganda National Panel Survey

This study uses the first five waves of the Uganda National Panel Survey (UNPS), a household survey carried out by the Uganda Bureau of Statistics as part of the World Bank LSMS-ISA project.[15] The UNPS interviews approximately 3,000 households per wave. The first wave started in 2009-10, and its initial sample was revisited for two consecutive years: 2010-11 and 2011-12. In the fourth wave (2013-14), one-third of the initial sample was refreshed. The fifth wave (2015-16) used the sample from 2013-14. The survey is conducted annually from September to August, involving two visits spaced approximately six months apart to effectively capture the agricultural outcomes associated with the two harvesting seasons.

From a macroeconomic perspective, the UNPS has two main strengths. First, it provides nationally representative panel household data from one of the poorest economies in the world. Second, it allows for the recovery of the consumption, income, and wealth dynamics of households, effectively capturing their entire budget constraints. Specifically, the survey enables the quantification of agricultural production and inputs, which constitute the main economic activity for most families in Uganda. However, measuring these variables in this context is complex. In the subsequent subsections, I describe the computation process for consumption, income (including agricultural revenues), and wealth.

[14] $\lambda_t(x,z,y_{na})$ is the proportion of agents at time t in state (x,z,y_{na}), with $x \in \mathcal{X} \equiv [0,+\infty)$, $z \in \mathcal{Z} \equiv \{z_l, z_{lm}, z_m, z_{mh}, z_h\}$ ($z_j \in \mathbb{R}_{++}$), and $y_{na} \in \mathcal{Y}_{na} \equiv \mathbb{R}_{++}$; and $\mathcal{S} := \mathcal{X} \times \mathcal{Z} \times \mathcal{Y}_{na}$.

[15] Uganda Bureau of Statistics. Uganda National Panel Survey (UNPS) 2009-2015. The Living Standards Measurement Study (LSMS) follows a standardized survey implemented by the World Bank to assess living standards across countries. The Integrated Survey on Agriculture (ISA) is specifically designed to capture agricultural production, wealth, inputs, and expenditures in Sub-Saharan African countries.

3.1 Consumption

The UNPS includes three sections for household consumption: food consumption (over the last seven days), non-food and non-durables consumption (over the last month), and durables consumption (over the last year). I compute the consumption value for each good or service by summing the values from purchases, home production, and in-kind and gift transfers. Then, I compute the household aggregate consumption as the sum of food and non-durable items consumption, excluding durable goods. To maintain data integrity, outliers are trimmed by excluding the bottom and top one percent of total household consumption for each wave.

3.2 Income

Measuring household income in developing countries is challenging; see Deaton (2005) and Deaton (2019) for useful discussions. First, a significant portion of the economy operates within the informal sector, leading to irregular periods of work and frequent in-kind payments. Second, most households are self-employed, making it challenging for them to report their earnings accurately both conceptually and practically. Third, many of these self-employed individuals are farmers, which presents further data measurement issues, especially when many farmers operate at the subsistence level. To overcome these challenges, the LSMS-ISA surveys query households about earnings derived from formal and informal labor services, non-agricultural business and trade operations revenues, and costs (LSMS section). Moreover, the LSMS-ISA surveys incorporate an exceptionally detailed agriculture questionnaire (ISA section).

3.2.1 Agricultural Income

With the agriculture questionnaire, I compute the agricultural revenues for each household and each crop by summing the earnings from sales plus the value of non-sold production, using district-level median consumption prices per crop kilogram. From the production side, households are questioned about the total production of the crop, as well as the production of the crop allocated to the market, own consumption, gifts to other households, storage, and animal feed. Like many surveys in developing countries, households can report crop production in various units, including non-standard units such as baskets, pails, or bunches. Additionally, conversion rates can vary across crops and seasons. For accurate conversion of crop production into kilograms, I use the median of the conversion rates to kilograms reported by households for each wave, season, crop, and unit combination.

Subsistence farming is prevalent in Uganda. On average, households devote only about 26% of their agricultural output to the market. Thus, understanding Ugandan households' livelihoods and measuring agricultural yields requires accounting for non-sold agricultural production correctly. However, as Deaton (2019) argues, setting prices to evaluate non-sold production in monetary terms is challenging. While prices at the gate (selling prices) are commonly used in the literature, they may underestimate non-sold production. This is because the items sold may only represent a portion of the harvested crops, and prices at the gate are typically measured after the harvest when supply is abundant and prices are low.

To address this issue, I leverage the household questionnaire, which captures food consumption in quantity and value. Following the approach in De Magalhães and Santaeulàlia-Llopis (2018), I use median consumption prices at the district level to evaluate unsold agricultural production. This procedure allows for better capturing the shadow value of not selling the crop: the cost if the household did not produce the crop and had to buy it in the local market. Agricultural revenues for each season and wave are typically trimmed at the top 2 percent of the distribution.[16]

3.2.2 Intermediates

Agricultural intermediates expenditure is computed as the sum of the costs of chemical fertilizer, organic fertilizers, pesticides and herbicides, seeds, hired labor, and transport. Since the data report the use of intermediates at the plot level, for multi-crop plots I assume that the intermediates are distributed uniformly across the crops on the plot. For chemical fertilizer, organic fertilizer, pesticides, and herbicides, the costs include total expenditure plus the value of the non-bought quantities using median prices recovered from the survey.

Agriculture in Uganda is characterized by a minimal usage of intermediates. Table 14 in the Appendix presents the share of farmers using intermediates and their average expenditure across the five waves of the UNPS. On average, farmers spend only \$78.14 per year on intermediates. Decomposing across crops, the average intermediates expenditure in high-yield, high-risk crops is \$57.76, while for low-yield, low-risk crops it is \$46.25. The use of modern agricultural intermediates in Uganda is extremely low, with only 4.7% of farmers using chemical fertilizer and 12.8% using pesticides.

3.2.3 Non-Agricultural Earnings

Non-agricultural earnings include business profits, formal and informal labor income, and livestock earnings. The household questionnaire collects detailed data on each non-farming business, including operative months, gross revenues, wage expenditures, raw materials costs, and other expenditures. Business profits are calculated as the difference between monthly gross revenues and costs, multiplied by the number of months the business operated. Similarly, labor earnings aggregate yearly payments for primary and potential secondary jobs for each household member, regardless of whether the job is formal or informal. Livestock profits are determined by summing net animal sales, meat, milk, and egg sales, and subtracting associated costs. The value of unsold production is estimated based on the consumption of self-produced livestock products reported in the household questionnaire's consumption section.

[16] Uganda has 135 districts. For crops without a district-level consumption price, such as sorghum and groundnuts, I use district-level selling median prices. If these prices are unavailable, I use regional-level prices, and if those are also unavailable, I use nationwide prices.

3.3 Wealth

The data equivalent of assets in the model is liquid wealth, which encompasses stored crops, livestock, household assets, rents, and transfers. Household assets include a wide range of items such as housing and other buildings, non-agricultural land, furniture, household appliances, electronic devices, jewelry, vehicles, and other assets. Rents and transfers consist of property income, dividends, interest from current accounts or bonds, remittances, inheritance, and alimony.

The value of stored crops is determined using district-level consumption prices, while the values of livestock and household assets are based on self-reported values if the household were to sell the asset at the current moment. While in the calibration wealth only includes liquid assets, for the empirical analysis, wealth encompasses all assets owned by the household, including also farm capital and land value.[17]

3.4 Consumption, Income, and Wealth Summary

Table 16 in the Appendix summarizes household consumption, income, and wealth for rural Uganda (columns 2-4) and the entire country (columns 5-7), highlighting three insights. First, income and consumption estimates closely align, indicating high data quality. Income estimates from surveys in developing countries typically lag behind consumption estimates for various reasons including those related to the difficulties of measuring agricultural income—see (Deaton, 2005). Second, Uganda remains a very poor economy. In rural areas, average household income and consumption are around \$1,505 annually, rising to \$1,736 nationwide. On a per capita basis, this equates to \$287 in rural areas and \$338 nationally. Third, while consumption inequality is comparatively low (Gini index of 0.33 respectively) income inequality levels are high with a Gini index of 0.57—compared to the United States. [18]

4 CALIBRATION

A set of parameters is taken from the literature or estimated directly from the UNPS data; the remaining ones—the technology factors A and B, the intermediates price p, the productivity dispersion σ_z, and the depreciation rate δ—are calibrated within the model so that its stationary equilibrium reproduces a set of time-averaged moments in rural Uganda. The calibration is a near-triangular method of moments: each calibrated parameter is pinned mainly by a single moment, while average income and the dispersion of wealth are left as over-identifying checks.

[17] To value land, in the first two waves, households reported their value, allowing for the computation of per-acre land prices grouped by counties and plot characteristics. For the other waves, land value is estimated by applying the county per-acre prices from 2010–11 according to plot characteristics.

[18] Values in 2013 US\$ not adjusted by PPP. Using 2006 PSID statistics, the Gini index of consumption in the United States is 0.41, while the Gini index in income and in wealth are 0.44 and 0.61, respectively—see De Magalhães and Santaeulàlia-Llopis (2018). Note that the UNPS has a relatively small sample and it does not over-sample the rich.

I compute the model's stationary distribution with the non-stochastic method of Young (2010), so the moments are evaluated deterministically and the mapping from parameters to moments is smooth—a method of moments rather than a simulated method of moments. Table 3 reports the targeted moments and the model's fit; below I describe how each parameter is chosen.

Preferences. The values of preference parameters are selected directly from the literature. I set the curvature of consumption, ρ, to two, a value commonly used in the macroeconomics literature—e.g., Kaplan and Violante (2010), Lagakos, Mobarak, and Waugh (2023)—and within the range of commonly micro-estimated values. For the discount factor, β, I set its value to 0.96, as in Aiyagari (1994), Donovan (2021), and other works.

Table 2: Parameter Values of the Model

	Parameter	Value	Source
Curvature of consumption	ρ	2	Literature
Discount factor	β	0.96	Literature
High technology inputs elasticity	α	0.4	Literature
Low technology inputs elasticity	γ	0.4	Literature
Intermediates price	p	30	Normalization
High-technology constant factor	A	275.73	Calibration
Low-technology constant factor	B	174.45	Calibration
SD permanent productivity	σ_z	0.2762	Calibration
Depreciation rate	δ	0.1110	Calibration
SD high-technology shock	σ_θ	1.0780	Calibration
SD low-technology shock	σ_ε	0.8264	Calibration
Covariance high and low shocks	$\sigma_{\theta,\varepsilon}$	0.1038	Calibration
Constant term non-agric earnings	$\bar{y}_{na}$	log(497)	Calibration
Persistence non-agric earnings	ρ_{na}	0.397	UNPS estimate
SD non-agricultural earnings	σ_{na}	1.335	UNPS estimate

Notes: A, B, $\bar{y}_{na}$, σ_z, δ calibrated targeting time-averaged cross-sectional moments. p is a normalization of the units of intermediate inputs. Risk parameters—σ_θ, σ_ε, $\sigma_{\theta,\varepsilon}$—calibrated targeting volatility measures of output residuals. ρ, β, α, and γ: see references in the text.

Technology parameters and inputs price. As is common in the literature, I set the intermediates elasticities α, γ equal to 0.4, the intermediates share in agriculture in the United States (Restuccia, Yang, and Zhu, 2008), and impose the same value on both technologies. I then calibrate the constant factors A and B to match the average output of high-yield and low-yield crops across waves: A is identified by high-crop output and B by low-crop output. The input price p is not calibrated. It normalizes the units in which intermediate inputs are measured and is fixed at 30: the survey records input expenditure, so (m, A, p) and $(\kappa m,\ A\kappa^{-\alpha},\ p/\kappa)$ are observationally equivalent for any $\kappa > 0$ and no moment identifies p separately.

With the elasticities fixed at the literature value, the model cannot match observed agricultural output and the low intermediate expenditure recorded in the UNPS at the same time: calibrating A and B to output implies model intermediate spending of \$211 and \$131 for high-and low-yield crops, against \$57 and \$44 in the data. Matching the observed share would require elasticities near zero, since the model's cost share is approximately α times a wedge below one, and that would remove the mechanism this paper studies. The friction cannot account for the gap: it is already in the model, and it lowers intermediate use rather than raising it. Intermediate use in Ugandan agriculture is genuinely low—only 4.7% of farmers use chemical fertilizer (Appendix Table 14)—and the survey misses part of what is used, for example, own-produced seed use is not recorded. The recalibration in the robustness section that instead matches the expenditure share lowers the productivity gain from 12.8% to 1.8%, which I report as a lower bound. The rise in intermediate use under complete markets should therefore be read as a proportional change from the model's own base, not as a level comparable to survey expenditure.

Agricultural shocks. A crucial aspect of quantifying the model is capturing agricultural risk across the two crop types. In the model, crop-output fluctuations are driven solely by household shocks and changes in intermediate use; in the data, production fluctuates for many additional reasons. To isolate the relevant risk I calibrate the covariance matrix of the agricultural shocks with the two-step procedure following Kaboski and Townsend (2011). First, I run an OLS regression of output on factors outside the model that can generate output variation, for the high-yield crops, equation (18), and for the low-yield crops, equation (19).

$$ln(y^h_{it}) = \phi\, ln(Z^h_{it}) + \beta X_{it} + \gamma F_i + \psi T_t + ln\theta_{it} \tag{18}$$

$$ln(y^l_{it}) = \phi\, ln(Z^l_{it}) + \beta X_{it} + \gamma F_i + \psi T_t + ln\varepsilon_{it} \tag{19}$$

y_{it} is crop output for household i in wave t; Z_{it} is a vector of household-crop-specific variables such as land size and labor hours; X_{it} is a vector of time-varying household characteristics (household size and the gender, education, age, and age squared of the head); F_i are household fixed effects and T_t wave fixed effects. Removing time effects is particularly relevant because the model has no aggregate shocks. Second, using the estimated residuals, I set the crop-risk parameters σ_θ, σ_ε, and $\sigma_{\theta\varepsilon}$ so that the model in stationarity replicates the volatility of the estimated residuals.[19]

Non-agricultural earnings process. For the non-agricultural earnings process, the OLS regression $ln\, y^{na}_{it} = \bar{y}_{na} + \rho_{na}\, ln\, y^{na}_{it-1} + u_{it}$, $u_{it} \sim N(0, \sigma^2_{na})$, delivers $\hat{\bar{y}}_{na} = 3.642$, $\hat{\rho}_{na} = 0.398$, and $\hat{\sigma}_{na} = 1.335$. To match average non-agricultural earnings in the data (\$502)—and hence total household income—I set $\bar{y}_{na}$ in the model to $\log(497)$. Notably, with the estimated $\hat{\rho}_{na}$ and $\hat{\sigma}_{na}$ together with the calibrated agricultural shocks, the model generates a household income volatility (0.617) very close to the data (0.602), an untargeted moment. I discretize the AR(1) process into a five-state Markov chain using the Rouwenhorst procedure.

Permanent component. To calibrate the permanent component of agricultural productivity,

[19]The residual volatilities are $\overline{Var_i(\widehat{ln\theta_{it}})} = 1.150$, $\overline{Var_i(\widehat{ln\varepsilon_{it}})} = 0.683$, and $\overline{Cov_i(\widehat{ln\theta_{it}}, \widehat{ln\varepsilon_{it}})} = 0.113$. Volatilities are measured as the average household time-variances (and covariance) of output in the model and of residual output in the data.

I follow Huggett and Parra (2010) and approximate z_i with five equally log-spaced points on $[-3\sigma_z, +3\sigma_z]$ with equal mass of households at each point. I then set σ_z to match the Gini of agricultural production. Because σ_z is calibrated after the agricultural shocks, it captures the dispersion in agricultural production not explained by risk.

Depreciation rate. The depreciation rate is set so that aggregate savings match average holdings of liquid wealth—crop storage, livestock, household assets, and other holdings (remittances, alimony, royalties, and other transfers). The calibrated value is $\delta = 0.111$, consistent with the substantial costs of saving in developing countries—for a summary see Karlan, Ratan, and Zinman (2014)—and close to the value in (Donovan, 2021).

Table 3 reports the targeted moments and their model counterparts. The model matches the agricultural targets essentially exactly—average high- and low-crop output, the output volatilities and their correlation, and the Gini of agricultural production (within 0.002)—and the average liquid wealth within about one percent. Average household income, which is not targeted directly but follows from the calibrated crop output and non-agricultural earnings, comes out within 0.2% of the data.

Table 3: Benchmark Economy: Targeted Moments

Description	Model	Data
Avg. high-yield crops output	658.3	658.4
Avg. low-yield crops output	347.8	348.2
Avg. income[a]	1,503	1,506
Avg. liquid wealth	1,862	1,858
High-yield crops volatility	1.150	1.150
Low-yield crops volatility	0.683	0.683
High–low-yield crops correlation	0.113	0.113
Gini agricultural output	0.624	0.624

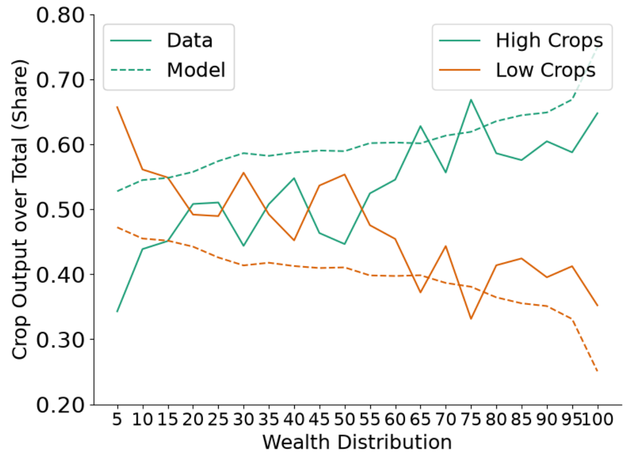
Notes: average and Gini moments are time-averaged across waves; volatility moments are computed on the residuals of $\ln(y_{it}) = \beta X_{it} + \gamma F_i + \psi T_t + u_{it}$. [a] Average income is not targeted directly (it follows from the calibrated crop output and non-agricultural earnings) and is reported as a check. *Source:* UNPS 2009/10–2015/16.

4.1 Model vs. Data

I evaluate the model against three sets of moments not used in the calibration: the crop portfolio along the wealth distribution, the joint distribution of consumption, income, and wealth, and the transmission of income volatility to consumption volatility.

Crop choice along wealth. Figure 5 plots the high- and low-crop output shares along the wealth distribution, in the data (solid) and the model (dotted). The model reproduces the key pattern—the high-crop share rises with wealth, increasing by 22% across the distribution against 26% in the data. It overstates the high-crop share among the poor (57% vs. 44% in the bottom quintile), a direct consequence of under-predicting wealth concentration (discussed below).

Figure 5: Crop Production along the Wealth Distribution: Model vs. Data



Notes: Dotted lines represent the share of output predicted by the model while solid lines the data counterparts. *Source:* UNPS 2009/10–2015/16.

Consumption, income, and wealth distributions. The calibration targets average income, average wealth, and the Gini of agricultural output (0.62), but no dispersion moment of the income, consumption, or wealth distributions. Table 4 compares the model in stationarity with the 2011–12 UNPS wave. The model tracks the income distribution well: the top half holds 84.9% of income in the model against 86.9% in the data, the top 5% 24.1% (28.4%), and the bottom 5% 0.36% (0.22%). It under-predicts the concentration of liquid wealth—the top 10% hold 37.6% versus 60.7% in the data, the top 1% 7.2% versus 21.7%—the standard limitation of incomplete-markets models, which lack the heterogeneous returns or discount rates, entrepreneurship, or other features that generate the wealth fat tail.

Most notably, the model reproduces the consumption distribution closely, with no consumption target. The top half holds 75.9% in the model against 73.8% in the data; the top 1%, 5%, and 10% shares (3.6%, 14.8%, 25.9%) nearly coincide with the data (3.8%, 14.7%, 25.1%); and the bottom matches as well (bottom 10% at 2.3% in the model, 2.8% in the data). Therefore, The model reproduces the low transmission of income inequality to consumption inequality that De Magalhães and Santaeulàlia-Llopis (2018) document for Sub-Saharan Africa.

Income and consumption volatility. The cross-sectional distribution shows how much income inequality is transmitted to consumption; its dynamic counterpart is how much idiosyncratic income risk passes through to consumption, which measures insurance directly. The model is calibrated to the volatility of high- and low-crop output and to an estimated AR(1) for non-agricultural earnings, and generates an idiosyncratic household income volatility of 0.617, close to the data (0.602). It reproduces the resulting consumption volatility—0.111 against 0.119 in the data (Table 17)—and hence how strongly income shocks are smoothed on their way to

Table 4: Income, Wealth, and Consumption Inequality. Model vs. Data

	Bottom					Top				
	1	**5**	**10** (%)	**25**	**50**	**50**	**25**	**10** (%)	**5**	**1**
Data										
Income	0.02	0.22	0.67	3.28	13.10	86.93	65.72	41.42	28.39	10.90
Wealth	0.01	0.10	0.29	1.43	6.13	93.88	80.55	60.67	47.46	21.73
Consumption	0.17	1.15	2.79	9.49	26.28	73.76	48.13	25.13	14.67	3.82
Model										
Income	0.04	0.36	1.02	4.11	15.11	84.89	63.88	37.83	24.09	8.75
Wealth	0.00	0.06	0.33	2.52	12.27	87.73	64.92	37.60	23.53	7.15
Consumption	0.11	0.91	2.28	8.37	24.14	75.86	50.41	25.95	14.76	3.59

Notes: Shares of the bottom percentiles groups, columns (1) to (5), and the top percentiles groups, columns (6) to (10) of the income, liquid wealth, and consumption distributions in the data and in the model. *Source:* UNPS 2011–12.

consumption.

That an incomplete-markets model compresses consumption relative to income is, on its own, expected (Aiyagari, 1994). Three features make the present exercise informative. First, it accounts for the income and consumption distribution in a Sub-Saharan setting, where structural accounts of the joint consumption–income–wealth distribution remain scarce. Second, the model captures the transmission of income to consumption volatility, and therefore the degree of consumption insurance observed in developing countries (Townsend, 1994; De Magalhães and Santaeulàlia-Llopis, 2018). It does so without risk sharing, through costly saving: the depreciating asset holds self-insurance to the degree seen in the data, with consumption volatility matching the data at the calibrated depreciation rate and falling well below it when saving is cheap (Figure 7). Third, it does so with a standard incomplete-markets model and literature-standard preferences ($\rho = 2$, $\beta = 0.96$, a no-borrowing constraint, and a depreciating asset), with no consumption target. What compresses the distribution further is the income process: a buffer stock smooths transitory shocks well and persistent ones poorly, and income here is less persistent than in calibrations of these models for high-income countries. Tractable incomplete-markets models are thus a useful and still underused instrument for low-income settings, where consumption data exist but heterogeneous-agent models are scarce.[20]

[20] Capturing the consumption insurance (low transmission income to consumption inequality and low transmission income to consumption volatility) is essential for assessing the effects of incomplete markets: the benchmark must reproduce the degree of consumption insurance, since it shapes both the size of the risk-induced distortions and their distributional consequences for welfare and crop choice.

5 Complete Markets in Agriculture: the Planner's Problem

I obtain the main quantitative results of this paper by comparing the previous benchmark economy to an economy with complete financial markets on agriculture. To solve this counterfactual world, I develop the following social planner's problem.[21]

The planner is able to allocate intermediates and consumption allocations from the total agricultural production. To formulate the problem, I follow the approach in L. Maliar and S. Maliar (2003), and I represent the planner's problem in the form of two sub-problems. The first sub-problem is to distribute aggregate consumption across heterogeneous agents, which delivers the social period utility

$$U(C_t) = \underset{c_{it}}{\text{Max}} \int_i \omega_i u(c_{it}) di \tag{20}$$

$$\text{subject to} \qquad \int_i c_{it} di = C_t \tag{21}$$

in which ω_i is the Pareto weight on individual i. The second sub-problem is to solve for the aggregate consumption that maximizes the social preferences

$$\underset{\{C_t, m_{it}^h, m_{it}^l\}_{t=0}^{\infty}}{\text{Max}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t) \tag{22}$$

$$\text{subject to} \quad C_t = \int_i \left[\theta_{it} z_i A (m_{it-1}^h)^{\alpha} + \varepsilon_{it} z_i B (m_{it-1}^l)^{\gamma}\right] di - p \int_i \left[m_{it}^h + m_{it}^l\right] di \tag{23}$$

$$C_t = Y_t - pM_t \tag{24}$$

in which aggregate consumption at period t is fully determined by the investment decisions on intermediates across individuals and technologies. Hence, the first-order conditions with respect to m_{it}^h and m_{it}^l are

$$\left[m_{it}^h\right] : \mathbb{E}_t \left[\beta^{t+1} U_c(C_{t+1}) \alpha \theta_{it+1} z_i A (m_{it}^h)^{\alpha-1}\right] = \beta^t u_c(C_t) p \tag{25}$$

$$\left[m_{it}^l\right] : \mathbb{E} \left[\beta^{t+1} U_c(C_{t+1}) \gamma \varepsilon_{it+1} z_i B (m_{it}^l)^{\gamma-1}\right] = \beta^t u_c(C_t) p \tag{26}$$

for all individuals $i \in \mathcal{I}$ at period of time $t > 0$. Given that θ_{it+1} and ε_{it+1} are independently distributed across time and unobservable by the planner, the intermediates allocations m_{it}^h, m_{it}^l are also independent of the idiosyncratic shocks realizations θ_{it+1} and ε_{it+1}. Also, note that in this economy, there are no aggregate shocks, and households have preferences for a smoothed consumption profile, so aggregate consumption growth needs to be constant. Thus, the expec-

[21] To replicate complete markets in agriculture, the planner provides full insurance for the agricultural output, but the planner cannot observe nor dictate on the realizations of the non-agricultural earnings.

tations on the first-order conditions simplify to

$$\left[m_{it}^{h}\right]: \mathbb{E}\left[\theta_{it+1}\right]\beta U_c(C_{t+1})\alpha z_i A(m_{it}^{h})^{\alpha-1} = u_c(C_t)p \tag{27}$$

$$\left[m_{it}^{l}\right]: \mathbb{E}\left[\varepsilon_{it+1}\right]\beta U_c(C_{t+1})\gamma z_i B(m_{it}^{l})^{\gamma-1} = u_c(C_t)p \tag{28}$$

in which $\mathbb{E}\left[\theta_{it+1}\right] = 1$ and $\mathbb{E}\left[\varepsilon_{it+1}\right] = 1$. Given that in the benchmark economy, aggregate consumption is stationary; I set the aggregate consumption from the planner solution to also be stationary, $C_t = C_{t+1}$. Thus, the first-order conditions become

$$\beta\alpha z_i A(m_{it}^{h})^{\alpha-1} = p \tag{29}$$

$$\beta\gamma z_i A(m_{it}^{l})^{\gamma-1} = p. \tag{30}$$

For optimality, the planner equalizes the discounted expected marginal product of each individual i on both technologies h, l to the marginal cost, the price. Input allocations are equal to expected discounted profit maximization. Isolating the input allocations from equations (29) and (30), we obtain the optimal levels of intermediate inputs

$$m_{h,i}^{*} = \left(\frac{p}{\beta\alpha A z_i}\right)^{\frac{1}{\alpha-1}} \text{ and } m_{l,i}^{*} = \left(\frac{p}{\beta\gamma B z_i}\right)^{\frac{1}{\gamma-1}}. \tag{31}$$

Moreover, note that conditions (29) and (30) imply that the marginal products across farmers (32) and technologies (33) equalize

$$\alpha z_i A(m_{it}^{h})^{\alpha-1} = \alpha z_j A(m_{jt}^{h})^{\alpha-1} \qquad \forall i, j \in \mathcal{Z} \tag{32}$$

$$\alpha z_i A(m_{it}^{h})^{\alpha-1} = \gamma z_i B(m_{it}^{l})^{\gamma-1}. \tag{33}$$

To solve the first sub-problem, I set the distribution of welfare weights in the planner's problem to replicate the competitive equilibrium under complete markets. In this regard, I set the Pareto weights such that each agent consumes according to its permanent component. That is, the shares of consumption across agents are equal to the expected shares of production. Thus, the consumption of an agent with productivity z_i is

$$c^{*}(z_i) = c_{it}(z_i) = z_i(A m_{h,i}^{*})^{\alpha} + z_i B(m_{l,i}^{*})^{\gamma} - p\left(m_{h,i}^{*} + m_{l,i}^{*}\right). \tag{34}$$

From the first sub-problem, note that the planner optimal consumption allocations will be such that the Pareto-weighted marginal utilities of consumption across agents equalize. As a consequence, the Pareto weights for agents i, j are set such that

$$\int_i \omega_i = 1 \qquad \frac{\omega_i}{\omega_j} = \frac{u^{-1}\left(c^{*}(z_j)\right)}{u^{-1}\left(c^{*}(z_i)\right)}. \tag{35}$$

The set of allocations from the planner provides the allocations of agricultural inputs and the consumption coming from its production under complete markets on agriculture. Given these allocations, households decide how much to consume and save out of the non-agricultural earnings process under the financial market of the benchmark economy.[22] This is by design: what the counterfactual targets is the insurance of agricultural risk alone. Completing all markets at once would also require a fully specified non-agricultural sector with extra assumptions difficult to verify in rural Uganda. This design is conservative: households still self-insure against non-agricultural risk, so precautionary saving remains and the gains reported below are smaller than completing all markets would deliver.

6 Quantitative Results

In this section, I present the numerical results of completing financial markets in agriculture. The first sub-section covers the gains in agricultural productivity and the risk-induced misallocation of intermediate inputs. The second covers the gains related to welfare: aggregate consumption, distributional effects, and an ex-ante welfare measure.

6.1 Agricultural Productivity

Completing financial markets leads to a 12.8% gain in agricultural output—and, equivalently, in agricultural productivity. Table 5 details the gains in output and input usage, both in aggregate and for the two crop types. Completing markets raises intermediate-input expenditure by 29.0%. In Sub-Saharan Africa, the low use of intermediate inputs is a factor behind the region's low agricultural productivity (Restuccia, Yang, and Zhu, 2008); uninsurable risk partially explains this low investment.

Table 5: Gains in Agricultural Output and Intermediates Usage

	Aggregate (%)	Low-yield Crops (%)	High-yield Crops (%)
Output	12.8	3.5	17.7
Intermediate inputs	29.0	6.5	43.0

Notes: proportional changes in agricultural output and intermediate expenditure from the benchmark economy in stationarity to the economy with complete financial markets in agriculture.

Uninsurable risk also distorts investment across crops. When financial markets are complete, expenditure on intermediates for low-yield crops rises by 6.5%, a 3.5% increase in output. High-yield crops see a much larger increase—43% in intermediate expenditure and 17.7% in output—so completing markets reallocates investment toward the higher-return crops.

Uninsurable risk also distorts investment among heterogeneous farmers. Under incomplete

[22]The Appendix A.1 states the household problem under complete financial markets in agriculture and provides the solution of the stationary equilibrium under this economy.

markets a farmer's investment depends not only on the permanent component but also on current income realizations and wealth holdings. Under complete markets it depends solely on the permanent component. Farmers with a low component reduce input usage by 2.8% and output by 0.3%, those with a middle component raise inputs by 17.7% and output by 8.3%, and those with a high component raise inputs by 34.9% and output by 14.2% (Table 18 in the Appendix reports the full breakdown, including the change within each farmer's crop portfolio).

The 12.8% therefore combines a higher level of intermediate use with a better placement of it, across crops and across farmers. The next two subsections quantifies these mechanisms. The first measures what the crop portfolio contributes, which works through both the level and the placement. The second holds the level fixed and measures the misallocation across farmers as well as crops.

6.1.1 The Contribution of the Crop Portfolio

To quantify the contribution of the crop portfolio, I build an economy with a single technology, $y_{it} = v_{it}\, z_i\, C\, m_{it-1}^{\alpha}$, and recalibrate it to the same aggregate moments as the benchmark: average agricultural output and the volatility of total agricultural output.[23]

The crop portfolio operates along three margins: diversification, the allocation of inputs across crops, and the level of input use. First, a single technology understates risk. The volatility of total agricultural output, a variance of 0.60, is what remains after households have diversified; the two technologies have 1.16 and 0.68 on their own. A single technology assigns that reduced volatility to the technology itself, so neither the diversification nor the return households gave up for it appears in models with a single technology. Second, with one technology there is no portfolio to allocate, and the misallocation of inputs across crops disappears. Third, input use is then distorted only once, by risk. In the benchmark it is distorted twice: directly by risk, and again through the shifting toward the low-yield technology, which lowers the marginal product of inputs. Completing markets undoes both, and intermediate use rises by 29.0% in the benchmark against 24.7% under a single technology.

Taking the three together, the crop portfolio accounts for 16.0% of the productivity gain: completing markets raises agricultural output by 10.7% in the single-technology economy, against 12.8% in the benchmark. This measures the crop portfolio as it operates through intermediate inputs. Farmers in the data also change their portfolio by reallocating land, and potentially labor and farm capital; the misallocation of those inputs across crops is outside of this work, and a fuller accounting of the crop margin would need them.[24]

[23] The technology parameters $\{A, B, \sigma_\theta, \sigma_\varepsilon, \sigma_{\theta\varepsilon}\}$ are replaced by $\{C, \sigma_v\}$; everything outside the production function—σ_z, δ, ρ, β, α and the input price p—is held at its benchmark value. Average liquid wealth and the Gini of agricultural output are not targeted and serve as checks; recalibrating δ and σ_z to restore them moves the contribution reported below by 0.15 percentage points.

[24] Grouping crops into two technologies rather than a finer classification works in the same direction. Splitting the data into three or five groups raises risk within a group by 3% and 8% relative to the two-group split (Section 1.1),

6.1.2 Risk-Induced Misallocation

To measure the cost of the misallocation of inputs across farmers and crops, I constrain the complete-markets solution to the benchmark's level of investment. I use the planner's framework to hold the investment shares across the crop portfolio and among heterogeneous farmers at their complete-markets values, with aggregate intermediate expenditure matching the benchmark economy. Holding those shares fixed equalizes the marginal products across crops and among farmers with different permanent components, satisfying the optimality conditions in (33) and (32). The counterfactual is therefore the most output obtainable from the inputs the economy already uses.

Reallocating both margins accounts for 15.8% of the total productivity gain; equivalently, risk-induced misallocation costs 2.1% of agricultural output. Reallocating across farmers alone accounts for 12.2% and across crops alone for 3.2%, the residual 0.4% being their interaction.

6.2 Consumption and Welfare

In the absence of complete financial markets, households mitigate agricultural risk through income choices and costly precautionary savings. Completing markets removes the need for both, producing a substantially larger impact on aggregate consumption than on income. Panel A of Table 6 compares household averages of agricultural output, income, assets, and consumption across the two economies. Completing markets raises agricultural output by 12.8%, which translates into an 8.6% rise in household income. Without complete markets, households accumulate wealth through costly means—livestock, crop storage, or hoarding cash; completing markets reduces these costly assets by 59%. Combined with the higher income, this delivers a 17.6% increase in aggregate consumption.

Panel B of Table 6 shows the impact on the income and consumption distributions: the percentage change for households in the bottom of the distribution (bottom 1%, 5%, 10%, 25%, 50%) and for the same groups at the top. Completing markets has theoretically ambiguous effects on inequality. Insurance is egalitarian: under complete markets consumption no longer depends on shock realizations, so—conditional on the permanent component—households with good or bad agricultural luck consume the same. Working against this are two forces: households with high permanent components z_i gain the most in output and are concentrated in the middle and top of the distribution, and the reduction in costly savings benefits the asset-rich top disproportionately.

For income, the inequality-enhancing forces dominate: the income Gini rises from 0.53 to 0.54, with the bottom 50% gaining 7.8% and the top 50% gaining 9.1%. For consumption, the egalitarian force roughly offsets them, leaving inequality essentially unchanged—the consumption Gini varies from 0.371 to 0.370. The consumption gains are large throughout but not monotone: they are largest in the lower tail, where insurance raises the worst outcomes (bot-

so the two technologies capture most of the scope for diversification.

tom 1% +26.6%, bottom 5% +17.9%), sizeable through the upper-middle (top 50% +18.3%, top 25% +17.9%), and smallest at the very top (top 1% +2.4%).

Table 6: Consumption and Welfare Gains from Completing Markets

A. Aggregate Gains	Incomplete Markets (Income = 100)	Complete Markets (Income = 100)	$\triangle_{CM}$ (%)
Agric. Output	66.7	75.3	12.8
Income	100.0	108.6	8.6
L. Wealth	121.8	49.8	-59.1
Consumption	63.1	74.2	17.6

B. Distributional Gains	Bottom					Top				
	1	**5**	**10** (%)	**25**	**50**	**50**	**25**	**10** (%)	**5**	**1**
Income	3.7	3.1	4.9	3.8	7.8	9.1	9.5	10.2	6.5	13.9
Consumption	26.6	17.9	16.8	14.8	15.5	18.3	17.9	10.0	6.2	2.4

Notes: Panel A normalizes all quantities by aggregate incomplete-markets income (set to 100); the aggregate resource constraint is $C = Y_a - pM + Y_{na} - \delta A$. Panel B reports the proportional change in total income (row 1) and consumption (row 2) of the bottom and top percentile groups. The income Gini rises from 0.53 to 0.54; the consumption Gini is essentially unchanged, from 0.371 to 0.370.

Consider the ex-ante welfare comparison of the benchmark and complete-markets economies. The average welfare gain, in consumption-equivalent variation (CEV), is the constant proportional increase in consumption g, common across time and agents, that equates utilitarian welfare across the two economies:

$$\int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u((1+g)c_{it}^{IM}) \right] d\lambda^* = \int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}^{CM}) \right] d\lambda^*. \tag{36}$$

Under the baseline calibration, the welfare gain g is equivalent to a 17.4% increase in lifetime consumption. The welfare gain is almost equal to the gain in aggregate consumption because the upper half of the distribution gains slightly more than the lower half, pushing the welfare gain below the consumption gain; while the fall in within-household consumption volatility pushes it up. Neither is large: the consumption Gini moves only from 0.371 to 0.370. The welfare gain mainly reflects the level of consumption rather than a change in how it is distributed or in how much it fluctuates.[25]

[25]Completing markets lowers the within-household variance of log consumption from 0.110 to 0.072, but raises the variance between households from 0.407 to 0.451: consumption no longer moves with agricultural shocks, but it depends fully on the permanent component. Cross-sectional dispersion is therefore almost unchanged. Because the welfare criterion below aggregates utilities across households and dates, it responds to that cross-sectional dispersion rather than to the within-household component separately, which is why the welfare gain tracks the rise in aggregate consumption.

6.3 Discussion and Robustness

Are these gains large? Set against the large cross-country agricultural productivity gaps and the sizeable misallocation shares documented in the macro-development literature, a 12.8% productivity gain is moderate. It is also concentrated: farmers with a low permanent component barely gain (Appendix Table 18). Full and efficient insurance does not make intrinsically unproductive farms—poor land, tiny plots, remote locations—productive. The productivity side is, moreover, conservative. I hold relative crop prices fixed—their adjustment could push the gains either way, and in any case most agricultural output in Uganda is not marketed. I hold the rural wage fixed, abstracting from cross-sector labor movement, and I abstract from land reallocation across farmers—both forces that would raise the gains and drive structural transformation. I abstract from these because they are hard to justify in an agrarian economy with incomplete land markets and subsistence-based production.

Yet, set against the high consumption insurance described in rural developing economies, the gains are large. Rural economies like Uganda appear, under the standard test that treats income as exogenous, close to fully insured, with a Townsend coefficient only about 0.04.[26] That near-complete smoothing, however, is achieved at a cost—the same incomplete markets distort households' income and investment choices enough to forgo the equivalent of 18% of consumption and welfare in this calibration.

Table 7 re-calibrates the model under alternative assumptions and reports its fit and the gains from completing markets.[27] The first alternative matches the low intermediate-expenditure share (0.10 in the data against 0.34 in the baseline), re-estimating the input elasticities α and γ; they fall to 0.11 and 0.14, well below the values estimated for agricultural production functions. This mechanically attributes the low input use to technology rather than to the friction, and the productivity gain collapses to 1.8%. The second drops bananas, addressing the concern that a single perennial crop drives the results. It does not: the risk trade-off between high- and low-yield crops increases, and the gains are not too different—10.3% in productivity, 12.0% in consumption, and 12.3% in welfare. The third recomputes the risk moments at the district level. Measuring risk from district-average yields removes household-level variation within districts—reducing measurement error but also removing household idiosyncratic risk. Total risk falls by roughly half, and the gains decrease but stay sizeable—5.6% in productivity, 7.1% in consumption, and 7.6% in welfare.

I also vary the depreciation rate $\delta \in [0, 0.15]$, the parameter that governs the cost of saving and hence the strength of the precautionary-saving channel. δ is calibrated to average liquid wealth; at the calibrated $\delta \approx 0.11$ the model also reproduces consumption volatility, which is

[26] Under CRRA preferences, the Townsend coefficient is the pass-through of idiosyncratic income into consumption, β, in $\Delta \ln c_{it} = \alpha \Delta \ln C_{rt} + \beta \Delta \ln y_{it} + u_{it}$, where aggregate regional consumption C_{rt} absorbs common shocks and I control for family size; $\beta = 0$ under full insurance. I estimate $\hat{\beta} = 0.0388$ (95% confidence interval [0.024, 0.054]).

[27] Each alternative re-fits the model—the scale parameters A and B, the shock variances, the permanent-productivity dispersion, and the depreciation rate δ—to match the crop yields, their risk moments, agricultural inequality, and average wealth, changing only the assumption noted.

not targeted, while wealth and consumption inequality are largely invariant to δ—consumption inequality close to the data throughout, wealth concentration below it throughout (Appendix Figure 7). Appendix Figure 8 plots the gains in agricultural productivity, consumption, and welfare across δ. The productivity gain does not substantially change on δ. It is 9.3% when saving is costless, 10.8% at $\delta = 0.025$, and between 12.2% and 12.6% for every δ at or above 0.05. Shutting down the precautionary-saving channel entirely therefore leaves close to three quarters of the productivity gain. The consumption and welfare gains behave differently, rising close to linearly—from 1.5% and 4.3% at $\delta = 0$ to 20.2% and 23.3% at $\delta = 0.15$.[28]

Table 7: Calibration Robustness: Model Fit and the Gains from Completing Markets

	Model				Quantitative Results		
	α/γ	input share —	$\sigma_\theta^2/\sigma_\varepsilon^2$	vol_c	Δ Ag. Prod. (%)	Δ Cons. (%)	Welfare (%)
Baseline	0.4	0.34	1.68	0.110	12.8	17.6	17.4
Match input shares	0.11/0.14	0.10	1.68	0.092	1.8	10.5	12.7
No bananas	0.4	0.34	1.73	0.100	10.3	12.0	12.3
District-level	0.4	0.36	2.61	0.083	5.6	7.1	7.6
δ sensitivity	0.4	0.34	1.68	0.04–0.13	9–13	1–20	4–23

Notes: "input share" is model $p(m_h + m_l)/(y_h + y_l)$ data is 0.10; $\sigma_\theta^2/\sigma_\varepsilon^2$ and vol_c are within-household variances of log output and log consumption. Match input shares, no bananas, and district-level are recalibration exercises while the delta sensitivity row is the range of each statistic as δ varies over $[0, 0.15]$.

7 Conclusion

This paper shows that uninsurable risk distorts intermediate-input investment and crop portfolios among heterogeneous farmers, reducing aggregate agricultural productivity and welfare. I begin by documenting three facts in the Uganda National Panel Survey: higher-return crops are also riskier, most farmers grow both high- and low-return crops, and wealthier farmers obtain most of their income from the high-return ones. These facts motivate an incomplete-markets model of crop-portfolio choice, which I calibrate to the same data and use to quantify the cost of missing insurance markets. Two results follow.

First, completing financial markets for agricultural risk raises output by 12.8%. The gain comes mainly from a higher level of intermediate use, a 29.0% rise, as insurance relieves the precautionary under-investment that keeps input use exceptionally low in the region; a smaller part, 15.8% of the total, comes from reallocating efficiently intermediates across crops and farmers. The crop portfolio contributes to both: an otherwise identical economy with a single tech-

[28] This complements the match-input-shares exercise: forcing a low elasticity there shuts down the income and investment channel and collapses the productivity gain, yet the consumption and welfare gains survive because they run largely through costly saving. With costless saving ($\delta = 0$) the consumption gain is small; the excess of the consumption and welfare gains over the output gain is the cost of the precautionary saving that completing markets eliminates, and δ measures how large that cost is.

nology yields a gain of 10.7% rather than 12.8%. Uninsurable risk is part of the explanation for the low intermediate use and the inefficient crop allocation observed in Sub-Saharan agriculture.

Second, the model reproduces the low transmission of income inequality and income volatility into consumption—the high consumption insurance in rural developing economies—yet shows that this smoothness is costly. Completing financial markets in agriculture delivers a welfare gain equivalent to a 17.4% increase in lifetime consumption. High consumption insurance and large welfare costs therefore coexist: the value of formal insurance here is less in protecting consumption, which is already smooth, than in removing the need to protect it through low-return production. The gains are uneven. They accrue to farmers whose potential is held back by risk, and little to those constrained by fundamentals—limited skills, small or remote plots. The results also offer some guidance for policy. Because households raise input use and shift toward higher-return crops on their own once risk is insured, financial products need not be tied to a productive use to raise agricultural incomes. For the same reason, a product that leaves consumption no smoother than informal arrangements already do need not be without value: the gain appears in production instead. And because the largest production gains come from farmers who are not the poorest, coverage may need to be relatively broad.

Data Source: Uganda Bureau of Statistics. Uganda National Panel Survey (UNPS) 2009–2016. Ref: UGA_2005-2009_UNPS_v02_M, ..., UGA_2015_UNPS_v02_M. Datasets downloaded from the World Bank Microdata Library, UNPS 2009, 2010, 2011, 2013, and 2015 on 31/05/2019.

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A Appendix

A.1 The Stationary Equilibria under Complete Financial Markets in Agriculture

The set of pareto optima allocations—in section 5—provides the allocations of agricultural inputs and the consumption coming from its production under complete markets on agriculture. Each agent i produces in agricultural outputs and receives the consumption from this production according to their ex-ante permanent component (z_i).

$$m^*_{h,i} = \left(\frac{p}{\beta\alpha A z_i}\right)^{\frac{1}{\alpha-1}} \text{ and } m^*_{l,i} = \left(\frac{p}{\beta\gamma B z_i}\right)^{\frac{1}{\gamma-1}} \tag{37}$$

$$c^*(z_i) = c_{it}(z_i) = z_i(Am^*_{h,i})^\alpha + z_i B(m^*_{l,i})^\gamma - p\left(m^*_{h,i} + m^*_{l,i}\right). \tag{38}$$

Given these allocations, households decide how much to consume and save out of the non-agricultural earnings process under the financial markets of the benchmark economy. That is they solve the following problem in stationarity

$$V(a,z,y_{na}) = \underset{a',c}{\text{Max}} \quad u\left(c^*(z_i) + c\right) + \beta\mathbb{E}\left[V(a',z,y'_{na})\right] \tag{39}$$

subject to

$$a' \geq \underline{a} \tag{40}$$

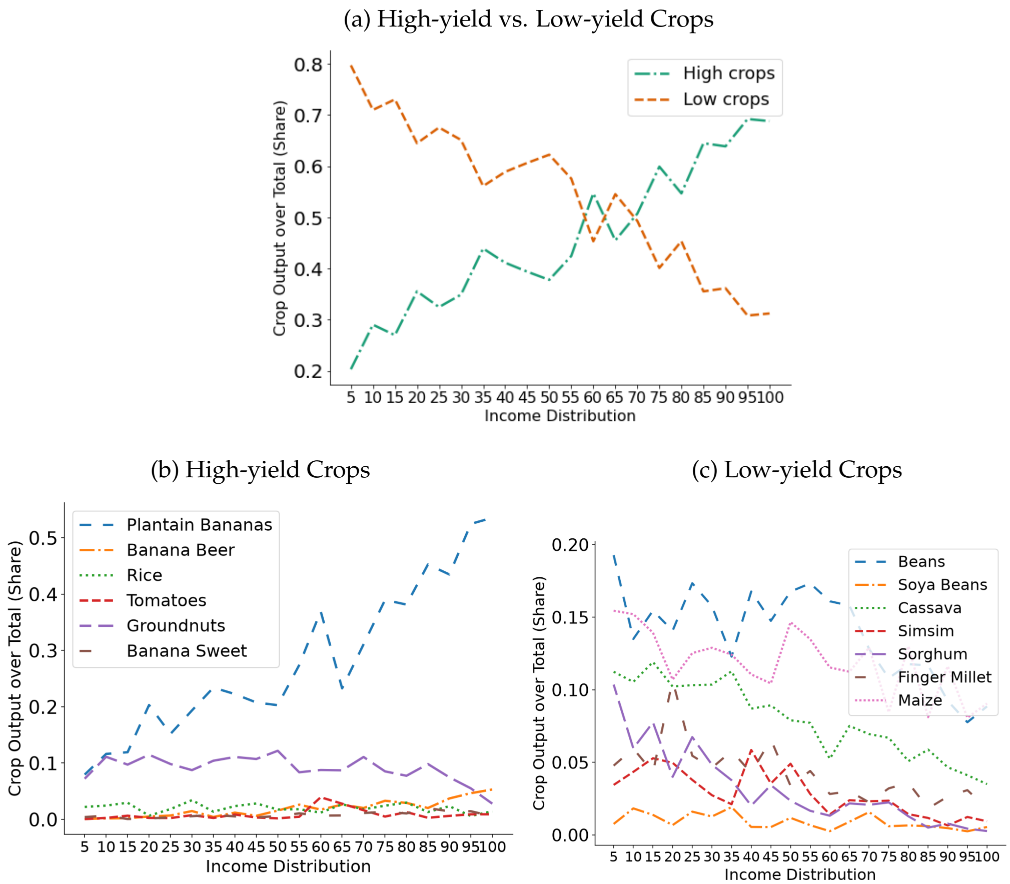
$$c(a,z,y_{na}) + a'(a,z,y_{na}) = (1-\delta)a + y_{na} \tag{41}$$

$$lny'_{na} = b + \rho_{na} lny_{na} + u, \qquad u \sim N(0,\sigma^2_{na}). \tag{42}$$

Finally, the definition of the stationary equilibria in the economy under complete markets on agriculture is analogous to the definition in section 2 with: the optimal allocations given by $m^*_h(z_i)$, $m^*_l(z_i)$, $c = c^*(z_i) + c(a,z,y_{na})$, $a'(a,z,y_{na})$; the transition function defined as $Q\left((a',z,y'_{na}),(a,z,y_{na})\right)$; the stationary distribution as $\lambda^*(a,z,y_{na})$; Under the space $\mathcal{S} := \mathcal{A} \times \mathcal{Z} \times \mathcal{Y}_{na}$ where $\mathcal{A} = [0,\infty]$.

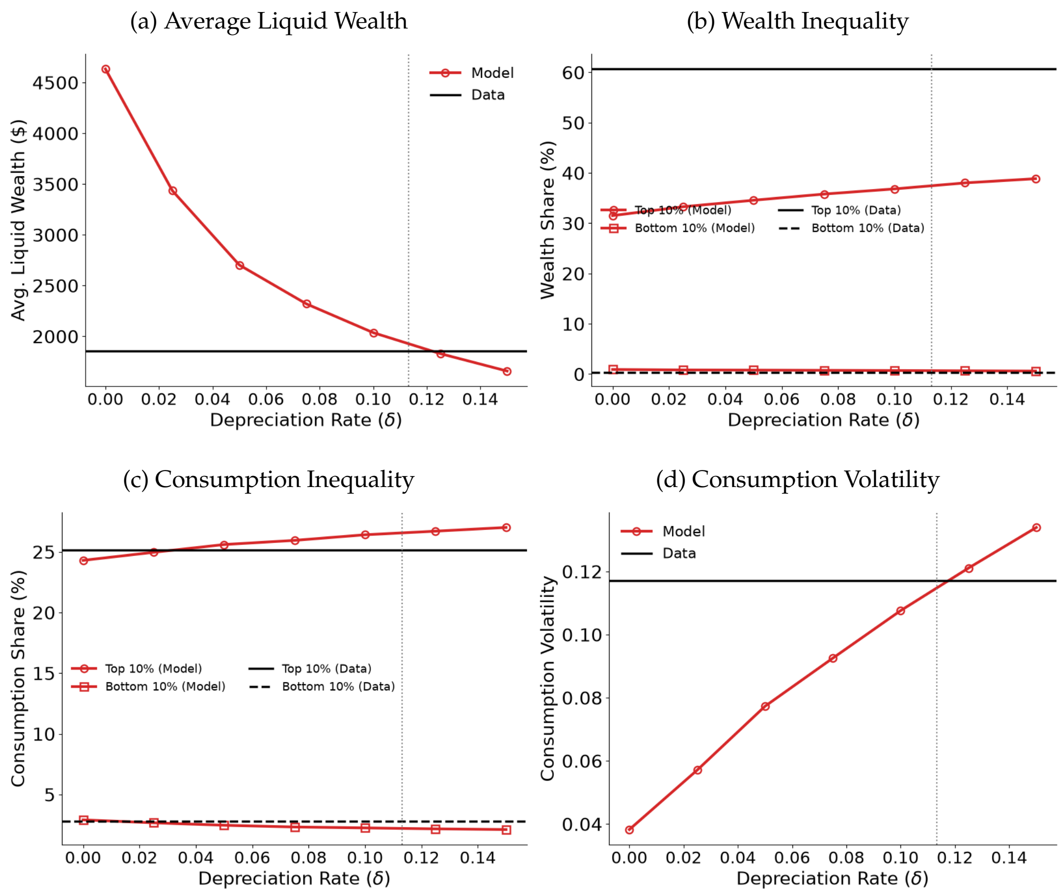
A.2 Tables and Figures

Figure 6: Crop Production along the Income Distribution



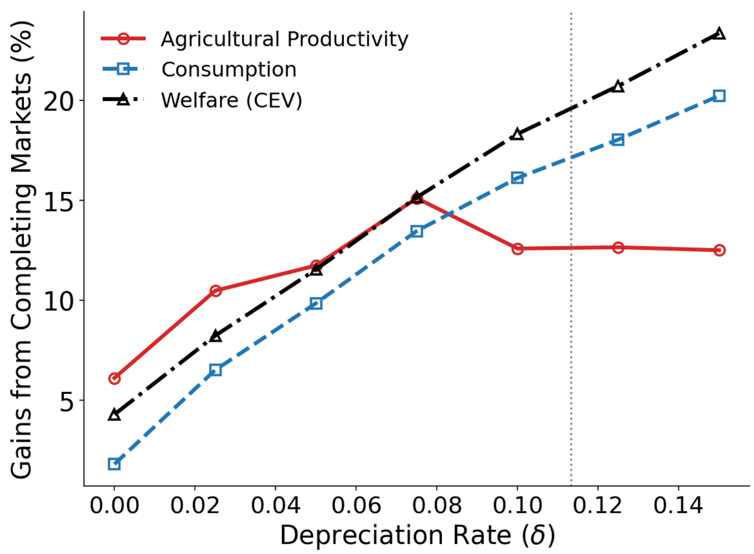
Notes: Crop production and income ranking computed from the average across the five waves. *Source:* UNPS 2009/10–2015/16.

Figure 7: Model vs. Data along the Depreciation Rate



Notes: each panel plots a model moment across the depreciation rate δ; red lines with markers are the model, black lines the data, and the vertical dotted gray line marks the calibrated δ used in the paper.

Figure 8: Gains Completing Markets along the Depreciation Rate



Notes: Gains completing financial markets in agriculture across the depreciation rate δ. Gray line is the calibrated δ.

Table 8: Crop Yields and Dispersion of the Yields

	N	Yield Avg	Risk Measures CV_{wave}	CV_{HH}	$SD_{\text{log resid}}$	CV_{dist}	CV_{kg}	$N_{dist.}$
	(N)	($/ac)		—				(N)
Corr(yield,risk)			0.684	0.443	0.373	0.678	0.220	
Crops								
Pumpkins	181	2,218	1.397	—	0.991	0.990	—	10
Eggplants	106	1,524	0.762	—	—	1.069	—	6
Yam	478	1,129	0.737	0.718	1.030	0.946	0.645	29
Rice	440	1,036	1.089	0.645	0.865	0.696	0.567	29
Cabbage	166	969	0.646	—	1.070	0.847	—	15
Sugarcane	332	946	1.257	0.813	1.015	1.032	0.809	19
Plantain Bananas	13559	891	0.611	0.826	1.102	0.864	0.654	71
Banana Sweet	925	805	0.640	0.830	0.935	0.906	0.619	38
Banana Beer	1437	731	0.494	0.870	1.154	0.949	0.685	35
Groundnuts	3813	679	0.499	0.768	0.993	0.778	0.671	84
Onions	166	516	0.458	0.653	1.180	0.977	0.762	12
Tomatoes	304	509	0.437	0.732	0.953	0.843	0.649	30
Pineapples	107	487	0.655	—	0.911	0.836	—	5
Irish Potatoes	1065	430	0.226	0.719	0.957	0.771	0.595	25
Cassava	6700	336	0.361	0.771	1.037	0.730	0.681	85
Simsim	950	302	0.538	0.743	0.721	0.835	0.530	26
Finger Millet	1994	290	0.192	0.675	0.857	0.661	0.644	57
Maize	15947	288	0.717	0.724	1.065	0.764	0.648	94
Beans	18131	250	0.235	0.733	1.019	0.598	0.632	90
Sweet Potatoes	6018	249	0.296	0.724	0.932	0.627	0.633	84
Soya Beans	828	225	0.918	0.662	0.791	0.845	0.605	29
Field Peas	443	198	0.501	0.791	0.927	0.747	0.772	17
Sorghum	2086	195	0.561	0.700	0.888	0.817	0.635	59
Pigeon Peas	460	182	0.415	0.638	0.717	0.611	0.546	16
Cow Peas	225	180	0.702	—	1.028	0.693	—	10
Cotton	351	165	0.345	0.641	0.756	0.765	0.638	15
Sun Flower	524	140	0.533	0.573	0.703	0.696	0.588	10

Notes: First row reports cross-crop correlation between average yield and each risk measure. Columns: (1) hh–wave observations; (2) average yield ($/acre); (3) CV of wave-level means (Figure 1 y-axis); (4) mean within-household CV across waves (min. 25 households); (5) mean within-household SD of log residuals (net of wave and season fixed effects); (6) mean within-district temporal CV; (7) mean within-household CV of physical yield (kg/acre); (8) number of districts. Dash indicates too few observations. *Source:* UNPS 2009/10–2015/16.

Table 9: Decomposition of Crop Yield Risk into Quantity and Price Components

Risk measure	Quantity (s_Q)	Price (s_P)	Covariance ($2\,s_{QP}$)	Crops (N)
Across waves	0.757	0.892	−0.649	25
Wtn-hh, wave + season FE	0.781	0.281	−0.063	27
Wtn-hh, + household FE	0.771	0.307	−0.078	27

Notes: The decomposition uses identity $\log(\text{value}/A) = \log(\text{kg}/A) + \log(p)$, where all output is valued at median district consumption price p, so that $\text{Var}(v) = \text{Var}(q) + \text{Var}(p) + 2\,\text{Cov}(q, p)$. "Across waves" decomposes the variance of crop-wave means (Y-axis in Figure 1); the lower two rows decompose the within-crop variance of household-level log yields after removing the indicated fixed effects. *Source:* UNPS 2009/10–2015/16.

Table 10: Quantity–Price Decomposition of Yield Risk, by Crop

Crop	Quantity (s_Q)	Price (s_P)	Covariance ($2\,s_{QP}$)
Pumpkins	0.797	0.189	0.014
Eggplants	0.377	0.707	−0.084
Yam	0.677	0.435	−0.112
Rice	0.879	0.162	−0.041
Cabbage	0.800	0.316	−0.116
Sugarcane	0.901	0.306	−0.207
Plantain Bananas	0.707	0.344	−0.051
Banana (Sweet)	0.755	0.403	−0.158
Banana (Beer)	0.754	0.296	−0.050
Groundnuts	0.744	0.276	−0.020
Onions	0.678	0.335	−0.013
Tomatoes	0.702	0.271	0.027
Pineapples	0.909	0.112	−0.021
Irish Potatoes	0.693	0.356	−0.049
Cassava	0.684	0.296	0.020
Simsim	0.698	0.293	0.009
Finger Millet	0.813	0.211	−0.024
Maize	0.873	0.193	−0.066
Beans	0.766	0.317	−0.083
Sweet Potatoes	0.811	0.288	−0.099
Soya Beans	0.890	0.199	−0.090
Field Peas	0.843	0.196	−0.040
Sorghum	0.772	0.260	−0.032
Pigeon Peas	0.784	0.278	−0.062
Cow Peas	0.781	0.335	−0.117
Cotton	1.028	0.120	−0.148
Sun Flower	0.981	0.099	−0.080
Average	0.781	0.281	−0.063

Notes: Within-crop decomposition of variance of household-level log yield after removing wave and season fixed effects, corresponding to the second row of Table 9. *Source:* UNPS 2009/10–2015/16.

Table 11: Banana Yield Risk by Proxying Stand Maturity

	Full Panel (2009–2015)			Common Window (2011-16)		
	Obs	Yield	CV_{HH}	N	Yield	CV_{HH}
	(N)	($/ac)	—	(N)	($/ac)	—
All banana growers	15,921	809	0.829	9,799	915	0.756
Established	12,127	691	0.879	6,005	742	0.792
Later adopters	3,794	1,189	0.713	3,794	1,189	0.713

Notes: Bananas are the plantain, sweet, and beer varieties pooled. A household is "established" if first observed growing bananas by 2010/11 and a "later adopter" if from 2011/12 onward; since plants take two to three years to bear, later adopters hold the younger stands. N is household–wave observations with positive yield, average yield is in $ per acre, and CV_{HH} is the mean within-household coefficient of variation of yield across waves. *Source:* UNPS 2009/10–2015/16.

Table 12: Alternative Crop Classifications

	Crops (H-L)	σ^2_θ	σ^2_ε	$\frac{\sigma^2_\theta}{\sigma^2_\varepsilon}$	High Share (Rich)
	(N)		—		(%)
Baseline (all crops)	14–14	1.178	0.686	1.716	55
Split median, $N \geq 150$	6–7	1.245	0.696	1.787	52.4
Split tercile	5–5	1.177	0.695	1.694	50
Excl. bananas	5–5	1.205	0.695	1.735	40.9
GAEZ physical	9–10	0.948	0.877	1.081	63.5
District-level	6–7	0.52	0.25	2.10	52.4

Notes: σ^2_θ and σ^2_ε are the mean across households of the within-household variance over time of the idiosyncratic log-yield residual of $\ln(y_{it}) = \beta X_{it} + \gamma F_i + \psi T_t + u_{it}$. for high- and low-yield crops, respectively; *High Share (Rich)* is the high-yield output share of the richest 20%. **Baseline.** High: Banana Beer, Banana Sweet, Cabbage, Eggplants, Groundnuts, Irish Potatoes, Onions, Pineapples, Plantain Bananas, Pumpkins, Rice, Sugarcane, Tomatoes, Yam. Low: Beans, Cassava, Cotton, Cowpeas, Dodo, Field Peas, Finger Millet, Maize, Pigeon Peas, Simsim, Sorghum, Soya Beans, Sunflower, Sweet Potatoes. **Split median ($N \geq 150$).** High: Banana Beer, Banana Sweet, Cassava, Groundnuts, Irish Potatoes, Plantain Bananas. Low: Beans, Finger Millet, Maize, Simsim, Sorghum, Soya Beans, Sweet Potatoes. **Split tercile.** High: Banana Beer, Banana Sweet, Groundnuts, Irish Potatoes, Plantain Bananas. Low: Beans, Maize, Sorghum, Soya Beans, Sweet Potatoes. **Excl. bananas.** High: Cassava, Finger Millet, Groundnuts, Irish Potatoes, Simsim. Low: Beans, Maize, Sorghum, Soya Beans, Sweet Potatoes. **GAEZ physical.** High: Banana Beer, Banana Sweet, Cassava, Irish Potatoes, Maize, Onions, Plantain Bananas, Sweet Potatoes, Yam. Low: Beans, Cabbage, Cowpeas, Finger Millet, Groundnuts, Pigeon Peas, Rice, Sorghum, Sunflower, Tomatoes. **District-level.** Crops are ranked and split by their district-level average yield, and $\sigma^2_\theta, \sigma^2_\varepsilon$ average households within each district–wave and then take the within-district variance over time of the residual, rather than the within-household variance; the resulting groups coincide with the split-median classification. *Source:* UNPS 2009/10–2015/16.

Table 13: Agricultural Production Allocations in Uganda

Wave	Share Production to: (%)				Proportion Farmers did: (%)			
	Sell	Cons	Stored	Gift	Sell	Cons	Stored	Gift
2009	25.0	57.0	4.4	5.8	72.7	93.2	35.0	57.6
2010	23.9	59.7	4.6	6.4	77.7	97.7	34.3	62.5
2011	24.8	59.2	4.6	6.3	78.6	98.8	33.1	63.9
2013	28.6	54.5	6.4	6.3	74.6	96.3	40.1	59.1
2015	29.4	55.7	6.8	5.2	77.0	98.8	44.3	57.4
Average	26.3	57.2	5.4	6.0	76.1	97.0	37.4	60.1

Notes: Values from all agricultural production, sample includes all crops grown in Uganda. "Share of Production" is the mean across households of the share of a household's total output (kg) allocated to each use; the remaining goes to seed, feed, on-farm use, etc. "Share of Farmers" is the percentage of households allocating more than 10 kg to each use. *Source:* UNPS 2009/10–2015/16.

Table 14: Intermediates Summary

A. Expenditure

	Total (m) ($)	High-Yield Crops (m_h) ($)	Low-Yield Crops (m_l) ($)
Avg	78.14	57.76	46.25

B. Composition

	Chem. Fert. (%) ($)	Org. Fert. (%) ($)	Herb-&-Pesticides (%) ($)	Seeds (%) ($)	Transport (%) ($)	Labor Pay (%) ($)
Use	4.7	13.2	12.8	61.9	12.9	46.6
Expend	56.90	75.60	45.05	21.91	14.91	66.23

Notes: Panel A shows the average expenditure in intermediates (2013 US$). The total is the average across all farmers, while the high- and low-yield columns average over the farmers growing crops of each type, so the two do not sum to the total. Panel B shows the proportion of farmers using each intermediate product (row Use, in per cent) and their average expenditure on it conditional on use (row Expend, in dollars). *Source:* UNPS 2009/10–2015/16.

Table 15: Farm Labor Summary: Household vs. Hired

Farm Labor Use	Value (%)
Household labor, share of total farm labor	92.2%
Hired labor, share of total farm labor	7.8%
Mean household labor days per season (days)	219.4
Mean hired labor days per season (days)	18.6
Household-wave hiring any labor	44.9%
Households hiring any labor at some point	65.7%

Notes: household-wave observations, labor days sum household and hired labor across plots and seasons. *Source:* UNPS 2009/10–2015/16.

Table 16: Consumption, Income, and Wealth in Uganda

A. Household Level

	Rural			National		
	Cons ($) ($[0,1]$)	Income ($) ($[0,1]$)	Wealth ($) ($[0,1]$)	Cons ($) ($[0,1]$)	Income ($) ($[0,1]$)	Wealth ($) ($[0,1]$)
2009-10	1,474.07	1,411.09	4,062.45	1,720.24	1,607.26	4,557.87
	(0.36)	(0.58)	(0.62)	(0.38)	(0.59)	(0.67)
2010-11	1,429.59	1,314.92	4,169.74	1,698.62	1,569.33	6,029.18
	(0.36)	(0.56)	(0.64)	(0.39)	(0.58)	(0.74)
2011-12	1,593.48	1,555.10	5,301.17	1,810.99	1,720.46	6,045.38
	(0.34)	(0.56)	(0.67)	(0.37)	(0.57)	(0.70)
2013-14	1,541.70	1,524.65	4,403.02	1,738.61	1,720.18	5,080.72
	(0.30)	(0.57)	(0.61)	(0.32)	(0.56)	(0.66)
2015-16	1,476.09	1,729.39	3,782.00	1,705.83	2,079.71	4,489.40
	(0.31)	(0.58)	(0.6)	(0.33)	(0.58)	(0.64)
Average	1,502.99	1,507.03	4,343.67	1,734.86	1,739.39	5,240.51
	(0.33)	(0.57)	(0.63)	(0.36)	(0.58)	(0.68)

B. Per Capita Level

	Rural				National			
	Cons ($) ($[0,1]$)	Income ($) ($[0,1]$)	Wealth ($) ($[0,1]$)	Hh Size (N) ($[0,1]$)	Cons ($) ($[0,1]$)	Income ($) ($[0,1]$)	Wealth ($)	Hh Size (N)
2009-10	265.77	261.05	732.80	6.57	321.13	303.45	764.83	6.41
	(0.36)	(0.6)	(0.63)	(0.26)	(0.41)	(0.61)	(0.66)	
2010-11	236.15	225.21	656.54	7.19	276.29	256.18	954.64	7.15
	(0.38)	(0.59)	(0.64)	(0.26)	(0.41)	(0.6)	(0.75)	
2011-12	258.41	252.23	814.90	7.55	291.19	277.69	867.85	7.52
	(0.38)	(0.59)	(0.69)	(0.26)	(0.41)	(0.59)	(0.7)	
2013-14	320.10	322.66	847.05	5.82	377.00	386.39	977.50	5.67
	(0.33)	(0.59)	(0.6)	(0.28)	(0.37)	(0.6)	(0.65)	
2015-16	326.26	402.70	838.02	5.65	395.71	495.39	939.35	5.52
	(0.35)	(0.6)	(0.6)	(0.28)	(0.4)	(0.63)	(0.65)	
Average	281.34	292.77	777.86	6.55	332.26	343.82	900.83	6.45
	(0.36)	(0.59)	(0.63)	(0.27)	(0.4)	(0.61)	(0.68)	

Notes: Panel A provides the household average (in 2013$) and Gini index within parenthesis while Panel B provides the same statistics but at per capita level—dividing household totals over household size. the measurements of consumption, income, and wealth are provided in the text. Wealth includes farm capital and land value. *Source:* UNPS 2009/10–2015/16.

Table 17: Income and Consumption Volatility. Model vs. Data

Description	Model Value (log)	Data Value (log)
Income volatility	0.617	0.602
Consumption volatility	0.111	0.119

Notes: Volatility moments are computed on the estimated residuals of regressions $ln(y_{it}) = \beta X_{it} + \gamma F_i + \psi T_t + u_{it}$. In which y is the variable for the volatility, X_t represents household characteristics, and F_i and T_t represent household and time fixed-effects. *Source:* UNPS 2009/10–2015/16.

Table 18: Agricultural Gains across Farmers' Permanent Component

	Input Usage			Output		
	Total (%)	High-Crops (%)	Low-Crops (%)	Total (%)	High-Crops (%)	Low-Crops (%)
Average	29.0	43.0	6.5	12.8	17.7	3.5
Permanent Component (z_i)						
Low	-2.8	2.7	-12.7	-0.3	1.7	-4.4
Mid-Low	7.0	15.2	-7.2	3.6	7.1	-3.3
Middle	17.7	29.1	-1.0	8.3	12.1	1.0
Mid-High	28.2	42.6	5.3	11.3	16.4	1.7
High	34.9	50.9	9.8	14.2	19.6	4.1

Notes: columns 1–3 report the proportional change in input expenditure and columns 4–6 the proportional change in output, from the incomplete- to the complete-markets allocation, for each group. The "Average" row reproduces Table 5.